

Survivorship Pension: Labor Supply and Optimal Insurance

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Abstract

In this paper, we separate labor supply responses to widowhood in the presence of survivorship pension into 1) response to widowhood, absent the pension, and 2) response to the benefit, within the widowhood state. We show that this distinction is critical for welfare analysis in a standard Bailey–Chetty formula. Using microdata from Brazil, we leverage both an event study around a household member’s death and two policy reforms that reduced pension duration and value. We find that benefit recipients reduce weekly hours worked by 22% four years after the spouse’s death, with no substitution to other labor market activities, such as self-employment or business ownership. Reductions in the duration and in the value of benefits reduce labor supply responses. We estimate an elasticity of Labor Earnings with respect to the lifetime benefit values of between 0.50 and 0.72.

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1 Introduction

The death of a primary income earner can be one of the most severe shocks a household can experience, posing significant economic risk for families by abruptly removing a core source of income and financial stability (Fadlon and Nielsen, 2021). In response, governments around the world have integrated survivor benefits into their broader social security systems as a key form of insurance against this risk. These programs are designed to replace lost earnings, prevent poverty among widows and dependents, and smooth consumption in the aftermath of bereavement (Coyne et al., 2024; Giupponi, 2019). Survivor pensions have thus become a central, if often under-examined, component of modern welfare states.

In this paper, we study how individuals’ labor supply responds to the death of relatives in the presence of survivors’ pensions, and how the duration and value of such benefits drive these responses. We explore the Brazilian pension system for dependents of deceased public sector and military workers, which provides a great empirical context to answer these questions. Brazil has one of the most generous and fiscally burdensome survivor pension systems in the world. Brazil’s government spending on survivor benefits is about 3% of GDP, the highest share among a large set of countries (roughly triple the OECD country average). Prior to recent reforms, surviving spouses have been entitled to receive up to 100% of the deceased worker’s pension or salary as a benefit, often without any conditions on the survivor’s age, income, or remarriage status.

We combine administrative data on the universe of pension recipients with the universe of the formal labor market, and use multiple complementary empirical strategies in our analysis. First, we implement an event-study design around the death of a household member in which we compare the labor outcomes of individuals before and after a relative’s death. Second, we explore a policy reform implemented in 2015 that reduced the duration of survivor benefits for younger individuals. This reform introduced sharp age-based cutoffs in pension entitlements: surviving spouses younger than 45 at the time of the death would henceforth receive a temporary pension (for a fixed number of years), whereas older survivors continued to receive lifetime benefits. Lastly, we exploit a second reform, implemented in 2019, that reduced the value of these benefits. Importantly, the second reform was not introduced to partners from military workers, providing us with an ideal control group to study the effects of benefit values on labor outcomes. We leverage both natural experiments and difference-in-differences designs, comparing labor supply responses between widowed spouses affected and those not.

We derive a sufficient-statistic model for optimal survivor insurance. The model trades off individuals’ willingness to pay for insurance with the fiscal externality induced by the

behavioral responses to the survivor benefit. We show that, even when individuals would like to fully smooth consumption across states, the government would not optimally choose to fully insure them. We show that the level of insurance depends on the labor-supply response to benefits, as well as the retirement response when marginal pension benefits do not perfectly match the contribution rate.

We then move to our empirical results. We begin by describing how labor supply changes with the death of a family member in the presence of survivors' pensions. First, we find that the event leads to a large reduction in labor supply by surviving family members. On average, we estimate that four years after the death of a household member, survivors who are entitled to a pension have reduced the weekly hours worked by about 22% and their labor earnings by about 13%. These are sizable behavioral responses, indicating that many survivors scale back or exit employment once the pension income begins. Notably, the effect is concentrated on the extensive margin of labor supply: we find a 16 p.p. reduction in the probability of employment 4 years after the event.

We subject the event-study findings to a variety of robustness and validation checks. Reassuringly, we find no evidence of differential pre-trends in labor supply prior to the death event. The reduction in employment emerges precisely in the period when survivor pensions commence, suggesting that it is the income from the pension, rather than the bereavement alone, driving the effect. We also show that the reduction in formal employment is not offset by transitions into other occupations. Using near-universe data on firm ownership, we find no increase in formal business activity and a decline in low-scale self-employment.

We next turn to understanding how the design of these benefits affects individuals' responses. We begin by studying the effects of benefit duration, focusing on a reform implemented in March 2015. Prior to the reform, all surviving spouses were entitled to lifetime pensions. The reform introduced an age-based scale: only spouses aged 44 or older retained lifetime benefits, while younger survivors received pensions for fixed durations ranging from 3 to 20 years. To recover the causal effects of the reform, we explore the age variation through a difference in differences design, comparing the labor supply of younger and older surviving spouses whose partners died before and after the reform.

We find that the duration of benefits had important effects on the labor supply of younger spouses. Those who faced a reduced duration in their benefit worked significantly more after the death of their partners. In turn, we do not find an anticipatory effect: the labor supply of spouses prior to their partners' deaths did not change because of the reform.

We then investigate how the value of survivor benefits affects labor supply responses.

To do so, we explore a second reform that substantially reduced the generosity of survivor pensions in Brazil’s public-sector pension system (RPPS). Prior to the reform, surviving dependents were generally entitled to receive 100 percent of the deceased worker’s pension benefit up to the social security ceiling, and 70 percent of the amount above the ceiling. Beginning in November 2019, the reform replaced this rule with a substantially less generous formula. Survivor pensions became equal to a base payment of 50 percent of the deceased worker’s pension benefit, plus an additional 10 percentage points for each eligible dependent, up to a maximum of 100 percent.

The 2019 reform applied to both the public-sector pension system (RPPS) and the general private-sector system (RGPS), but did not affect survivor pensions for spouses of deceased military workers, whose benefit rules remained substantially unchanged during this period. This institutional distinction provides us with a natural comparison group to study the effects of benefit generosity on labor supply. We recover the causal effects of the reform using a difference-in-differences design that compares the labor supply of surviving spouses of public servants and military workers whose partners died before and after the implementation of the reform. Intuitively, spouses of military workers capture changes in labor market conditions and widowhood responses unrelated to the reform, allowing us to isolate the effect of the reduction in survivor pension generosity on labor supply decisions.

We show that the reform reduced by, on average, 33% the benefit that spouses received. This translates to significant changes in the labor supply responses of individuals after the death of their spouses. Those who had their benefits reduced have a smaller reduction in labor supply than unaffected spouses. Once again, we do not see effects in labor supply prior to the death of their partners.

In our last section, we bring together our empirical estimates into the sufficient statistic framework. First, we translate our behavioral responses to the widowhood event and to the change in survivor benefits into interpretable elasticities. This requires translating the policy variation in benefits into changes in lifetime wealth, which allows us to compare the 2015 reform of benefit duration with the 2019 reform of per-period amounts. We show that the labor supply response to changes in lifetime wealth is consistent with other papers in the literature. Second, we show that, regardless of the curvature of the utility function, individuals are over-insured in the Brazilian context, as the labor supply elasticity with respect to the widowhood event is negative at the level of benefits prior to 2015. However, this does not imply that the optimal amount of benefit is zero. We derive an expression for the optimal benefit depending on the curvature of the utility function and the estimated fiscal externality.

Related Literature: This paper contributes to two often interconnected strands of literature. First, we contribute to the literature that studies labor supply responses to fatal shocks of family members. With a similar event-study design, [Fadlon and Nielsen \(2021\)](#) find that in Denmark, in the absence of more robust social insurance for disabilities and deaths, surviving household members increase their labor supply to compensate for negative shocks. [Coyne et al. \(2024\)](#) show that in the U.S., widows between 50-70 have very small, but positive labor supply responses after the spouse’s death.

Our findings contribute to this literature in different ways. First, we show that in the presence of generous survivors’ pensions, widows substantially reduce their labor supply in response to the death of their spouses. This is the opposite of what has been documented in Denmark and the U.S.. We also show that this response is the same if it is a Daughter receiving the benefit, which, to our knowledge, had not been documented in the literature before. We interpret these differences as evidence of the effects of survivors’ pensions in determining labor supply responses to the death of family members.

Second, we contribute to the literature on the design of such benefits. [Giupponi \(2019\)](#) studies a change in the amount received by the surviving household member in Italy and shows that widows almost fully offset the pension cut by increasing their labor earnings one-for-one, implying a nearly full income effect. Meanwhile, [Coyne et al. \(2024\)](#) focus on the eligibility threshold of 60 years old to receive benefits in the United States. They find that eligibility for survivors’ pensions causes a 3 p.p. reduction in labor force participation for widows at the age of 60. Similar effects are found in the Netherlands by [Rabaté and Tréguier \(2024\)](#) who explore a reform in eligibility to benefits.

2 Institutional Background and Data

Brazil’s social protection system is divided between non-contributory and contributory programs. Non-contributory benefits include universal healthcare through the *Sistema Único de Saúde* (SUS), the *Benefício de Prestação Continuada* (BPC), and *Bolsa Família*. Contributory programs include retirement pensions, survivor pensions, disability insurance, and sickness benefits tied to prior labor market contributions. These contributory benefits account for the majority of social protection spending in Brazil, representing roughly 70 to 75 percent of total expenditures when healthcare is included, and close to 90 percent when healthcare is excluded.

Contributory benefits are further divided across systems based on workers’ occupational status. Private sector workers, employers, and most self-employed workers are covered by the *Regime Geral de Previdência Social* (RGPS), while civil servants are covered

by the *Regime Próprio de Previdência Social* (RPPS). Members of the armed forces, military police, and firefighters are covered under separate military pension systems. While these systems differ in financing rules and benefit formulas, they all provide survivor pensions to dependents of deceased contributors. Survivor pensions have historically been one of the largest and most generous components of Brazil’s contributory system, particularly for public sector and military workers, making them a useful setting to study how changes in benefit generosity affect household behavior.

Spending on social security constitutes a major share of Brazil’s public expenditures. In 2023, total pension spending (across RGPS, RPPS, and military systems) reached over 13% of GDP, with the General Regime (RGPS) alone posting a deficit of R\$ 304.6 billion (2.5% of GDP).¹ The public sector pension system (RPPS), despite covering less than 10% of all beneficiaries, accounts for a disproportionately large share of total pension outlays. The military pension system is even more fiscally unbalanced: in 2023, its deficit reached R\$ 49.7 billion, driven in part by more generous eligibility rules and benefit formulas. Within this broader structure, survivor pensions represent a significant and politically sensitive component, especially in the RPPS and military regimes, where legacy entitlements and institutional protections have made reforms more challenging.

The relevance of survivors’ pensions in Brazil can be illustrated in a comparison with other countries. In Figure 1, we show expenditures in survivors’ benefits as a share of GDP. We observe that Brazil has the highest share when compared to a set of OECD countries, rivaled only by Italy and Spain, despite having a much smaller GDP per capita.

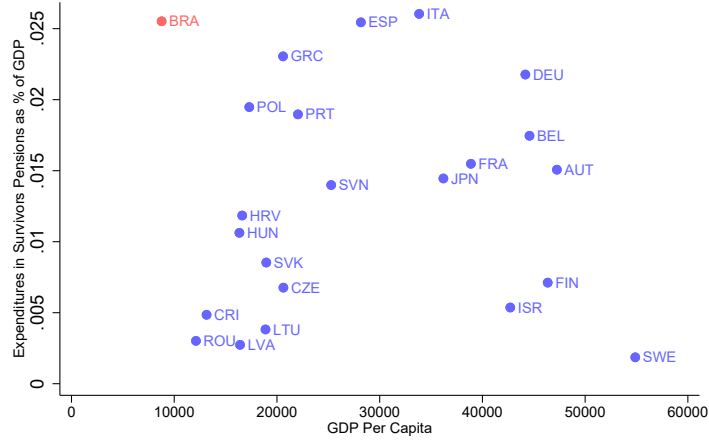
2.1 Survivor Pensions

Survivors’ pensions have been a longstanding feature of Brazil’s social protection system, especially for public sector and military workers. They were among the earliest formal social protection policies created by the Brazilian state. In one of the first sets of policies after the Proclamation of the Republic in 1889, the state created in 1890 the *Montepio Geral dos Servidores do Estado*, which functioned as a pension fund that provided support to widows and orphans of public employees. These benefits were designed to ensure that the death of a worker would not result in the immediate loss of household income, in a context where women’s labor force participation was substantially lower and household income often depended primarily on male earners.

Over time, these systems were formalized and absorbed into the state’s social security apparatus. Today, the rules governing survivors’ pensions vary significantly depending on the occupational regime of the deceased worker. Each regime has distinct eligibility

¹Ministério da Previdência Social (2024).

Figure 1: Expenditures in Survivors Benefits as a % of GDP



This Figure plots the share of government expenditures in survivors' benefits as a % of GDP on the y-axis and GDP per Capita in the x-axis. Data is collected from OECD, CEPAL, and World Bank country-level datasets. The sample comprises countries with available information, with expenditures bigger than 0.001%, and GDP per capita smaller than 60 thousand dollars (2015 US\$ measures)

criteria, benefit formulas, financing rules, and indexation mechanisms, but RGPS and RPPS converged and have similar survivor pension schemes. In this paper, we focus on the RPPS and military pensions because of data availability. Next, we outline the key institutional features that shape how survivors' pensions are granted and distributed to dependents of deceased public servants.

2.1.A Federal Workers Pensions (RPPS)

Next, we detail the rules of survivors' pensions under the RPPS system. We focus on the rules prior to the 2015 reform. We describe the important changes that took place in 2015 and 2019 below.

Eligibility: All spouses of deceased public servants or retirees are entitled to survivor pensions as long as the deceased individual had contributed at least 18 months to the public system. In addition to spouses, in some cases, other dependents are also entitled to the pension. When multiple dependents are eligible simultaneously (e.g., a surviving spouse and minor children), the pension is equally divided among all beneficiaries in the same priority group. Brazil's dependency hierarchy follows a predefined order: (1) spouse/partner and children; (2) parents; (3) siblings. Only one group can receive the benefit at a time. For example, if a spouse and two minor children are eligible, each will receive one-third of the total pension. As dependents lose eligibility (e.g., children reaching 21, or remarriage,

depending on local rules), their shares are redistributed among the remaining eligible dependents within the group. If no eligible members remain in the first group, the benefit may pass to the next group in the hierarchy, provided economic dependence is proven.

Benefit Base: The value of the pension is calculated based on the retirement benefit the deceased was receiving, or would have received if retired on the date of death. For active public servants, this corresponds to the integral or proportional value of the retirement benefit to which they would have been entitled, based on years of contribution and salary history. For retired civil servants, the pension is typically based on the gross value of the retirement benefit.

Replacement Value: Until 2019, the standard rule stipulated that 100% of the benefit was paid up to an yearly updated ceiling², and 70% of the marginal amount that exceeded the ceiling. The pension is nominally indexed, but indexation rules can vary and are subject to changes in legislation or subnational fiscal capacity.

Pension Duration: Until 2015, spouses received lifetime survivors' pensions. Children would receive until completing 21 years of age, or 24, depending on college enrollment.

2.1.B Changes in Federal Workers' Pension System

There were two large recent changes in the Federal workers' pension system, which we explore to understand the effects of the design of survivors' pensions. We explain them in more detail in sections 5 and 6.

The first major reform occurred in 2015, and changed the duration of benefits received by surviving spouses. Prior to the reform, spouses were generally entitled to lifetime survivor pensions regardless of age. The reform introduced an age-based schedule that sharply reduced benefit duration for younger widows and widowers: surviving spouses younger than 21 became eligible for only 3 years of benefits; those aged 21–26 for 6 years; 27–29 for 10 years; 30–40 for 15 years; and 41–43 for 20 years, while only spouses aged 44 or older remained eligible for lifetime pensions.

In 2019, Brazil enacted a second major reform that reduced the generosity of survivor pensions for new beneficiaries. The reform replaced the previous rule with a new formula that pays 50% of the benefit base plus an additional 10 percentage points for each eligible dependent, up to a maximum of 100%.

²The ceiling corresponds to the private system pension system ceiling, which is updated annually according to inflation and varies around 5.5 minimum wages. In 2024, the RGPS ceiling was R\$ 7786.02

2.1.C Military Pensions

Military pensions in Brazil follow a distinct set of rules established under the military social security system, which applies to members of the Armed Forces and operates separately from both the RGPS and RPPS. While also aimed at providing financial support to survivors of deceased personnel, military pensions have traditionally been more generous and less restrictive in eligibility.

In contrast to RPPS rules, military pensions are not subject to contribution-based ceilings, and the full value of the salary or retirement benefit is typically passed on to dependents. The pension is nominally indexed and preserved in real terms, though indexation mechanisms are defined by specific statutes applicable to military personnel.

2.2 Data

In this section, we describe the three main data sets used in our empirical analysis. The first set of data comprises the universe of survivors' pensions conceded to relatives of public sector and military workers. We then describe our labor market data that encompasses the universe of formally employed individuals in Brazil.

Pensionistas do Executivo Federal: This is a publicly available dataset that includes all recipients of survivors benefits originating from former Federal public sector workers between 1994 and 2025. On the recipient side, the data contains their full name, six digits of CPF (the Brazilian equivalent of social security number), date of birth, and how they relate to the deceased public servant. It also contains the dates of start and end of the pension (in case it is temporary), and the total gross and net payments that the individual received each month. Furthermore, it informs the full name, date of birth, date of death, and six digits of the CPF from the public sector worker who originated the pension.

Pensionistas Militares: This is also a publicly available dataset, which includes all recipients of survivors benefits but originating from former military workers. It contains similar information to the one described above, with the exception of the date of death of the former military employee and the date of birth of the recipient. It also contains the full name and six digits of CPF, which allows us to identify recipients in the labor market data.

Relação Anual de Informações Sociais (henceforth RAIS). This is the matched employer-employee data set that covers the universe of formal sector employees in Brazil. This data is organized by the Brazilian Ministry of Labor from information that firms submit annually. Firms have to inform all employees on their payroll for the respective year and provide information on their wages, hiring and separation dates, and other ob-

servable characteristics such as gender, age, and race.³ Importantly, since 2003, RAIS has provided information on names and the full CPF number of all workers, which is key to matching pension recipients to RAIS, in the procedure we detail below.

3 Conceptual Framework

This section develops a simple model for optimal survivorship insurance. Different from unemployment insurance, we assume that individuals' actions cannot change the likelihood of the bad state (widowhood). This means that there is no standard moral hazard problem as in other insurance problems. However, a trade-off persists, as survivors' benefits change at the margin, labor supply decisions within the widowhood state. This trade-off can be more easily conceptualized by separating the problem into two parts. On the one side, for a fixed amount of survivors' benefits, the widow's change in the marginal utility of consumption informs the need to insure across states. For example, if the marginal utility of consumption increases when switching to the widowhood state, this is informative that individuals would be willing to pay for insurance. On the other hand, within the widowhood state, an increase in the amount of benefits affects the marginal incentive to work and retire, reducing tax collection, and individuals' future retirement benefits.⁴

We consider a two-period model where surviving spouses can work in the first period, interpreted as working life, and then retire in the second period, interpreted as retirement. Individuals make two labor supply decisions: they choose their earnings when working, denoted by z , and they choose the age at which they retire, denoted by t_r . Time t follows the age of the survivor.

We make a key simplifying assumption: survivors perfectly smooth consumption over time after the death of their spouse. Under this assumption, the spouse's lifetime wealth at the spouse's death is a sufficient statistic for future consumption. This implies that the expected lifetime value of survivor and pension benefits matters rather than the per-period benefit amounts or benefit duration.⁵ This also implies that the model makes no meaningful distinction between being inactive and retiring, which helps connect the theory to the empirics.⁶ Another simplifying assumption is that survivor insurance benefits are

³The Ministry of Labor later uses this information to hand out benefits and tax exemptions for workers; thus, filling in RAIS is a high-stakes task for firms. Firms that do not submit the information to the Ministry of Labor are also subject to large penalties, and for all accounts, compliance is high.

⁴If the marginal benefit does not exactly match the social security contribution rate, which it does not for most cases, it creates a fiscal externality on top of the income tax.

⁵We come back to this when discussing the two reforms which induce variation in both per-period amounts and duration, separately.

⁶In practice, a key difference between being inactive and being retired is that individuals would

not means-tested: they do not depend on the survivor's earnings during working life or on the survivor's pension benefits in retirement.

3.1 Survivor's Problem and Behavior

Let $W(t_s, t_r, t_d, z)$ denote lifetime wealth as a function of the survivor's age at the death of their spouse, t_s , the survivor's retirement age, t_r , the survivor's age at death, t_d , and the survivor's earnings when working, z , with

$$t_s \leq t_r \leq t_d.$$

The survivor chooses working-life consumption c_w , retirement consumption c_r , retirement age t_r , and earnings z by solving

$$\max_{c_w, c_r, t_r, z} (t_r - t_s) \left[u_w(c_w) - \chi\left(\frac{z}{\omega}\right) \right] - \int_{t_s}^{t_r} \phi(t) dt + (t_d - t_r) \beta u_r(c_r) \quad (1)$$

$$\text{s.t.} \quad (t_r - t_s)c_w + (t_d - t_r)c_r \leq W(t_s, t_r, t_d, z), \quad (2)$$

where β is the discount factor applied to retirement utility, $u_w(\cdot)$ and $u_r(\cdot)$ are utility from consumption during working life and retirement, respectively, $\chi(\cdot)$ is the variable cost of effort required to earn z for an individual with earnings ability ω , and $\phi(t)$ is the fixed cost of working at age t .

Consumption decisions (c_w, c_r) solve

$$u'_w(c_w) = \beta u'_r(c_r), \quad (3)$$

$$(t_r - t_s)c_w + (t_d - t_r)c_r = W(t_s, t_r, t_d, z). \quad (4)$$

Therefore, consumption depends only on preferences, the discount factor, and lifetime wealth.

We assume that lifetime wealth is given by

$$W(t_s, t_r, t_d, z) = (t_r - t_s) [z - T(z)] + R(z, t_r) + B(y, t_s, t_d), \quad (5)$$

where $T(z)$ is the per-period tax-transfer amount on earnings during working life, $R(z, t_r)$ is the after-tax lifetime value of pension benefits received in retirement, and $B(y, t_s, t_d)$ is

typically receive pension benefits if they are retired, but not if they are inactive. Since per-period benefit amounts do not matter and only the expected lifetime value of benefits does, this difference becomes irrelevant. This helps tie the theory to the data, as we do not observe retirement decisions or retirement benefits, or observe them only imperfectly, in the matched employer-employee data.

the after-tax lifetime value of survivor insurance benefits. Survivor insurance benefits are allowed to depend on the deceased spouse's reference earnings, y , and on the survivor's age at the time of death, t_s , but not on the survivor's earnings z or retirement age t_r .

Earnings decisions solve

$$(t_r - t_s) \frac{1}{\omega} \chi' \left(\frac{z}{\omega} \right) = u'_w(c_w) \frac{\partial W}{\partial z} \quad (6)$$

$$= u'_w(c_w) \left[(t_r - t_s) (1 - T'(z)) + R'_z(z, t_r) \right]. \quad (7)$$

Retirement decisions solve

$$u'_w(c_w) \left[\frac{\partial W}{\partial t_r} - (c_w - c_r) \right] + [u_w(c_w) - \beta u_r(c_r)] = \phi(t_r) + \chi \left(\frac{z}{\omega} \right) \quad (8)$$

$$\Leftrightarrow u'_w(c_w) [z - T(z) - (c_w - c_r) + R'_{t_r}(z, t_r)] + [u_w(c_w) - \beta u_r(c_r)] = \phi(t_r) + \chi \left(\frac{z}{\omega} \right). \quad (9)$$

3.2 Government's Problem and Revenue-Neutral Reforms

For any potential deceased spouse earnings and survivor age at death, (y, t_s) , the government chooses the contribution to survivor insurance, $\tau(y)$, and survivor insurance benefits, $B(y, t_s, t_d)$, solving

$$\begin{aligned} \max_{\tau(y), B(y, t_s, t_d)} & \quad (1 - p(y, t_s)) \left[u_g(y - \tau(y) + z_s - T(z_s)) - \chi_d \left(\frac{z_d}{\omega_d} \right) - \chi_s \left(\frac{z_s}{\omega_s} \right) \right] \\ & + p(y, t_s) \left[(t_r - t_s) \left(u_w(c_w) - \chi \left(\frac{z}{\omega} \right) \right) - \int_{t_s}^{t_r} \phi(t) dt + (t_d - t_r) \beta u_r(c_r) \right] \end{aligned} \quad (10)$$

$$\text{s.t.} \quad (1 - p(y, t_s)) \tau(y) + p(y, t_s) [(t_r - t_s) T(z) - R(z, t_r) - B(y, t_s, t_d)] = E(y, t_s), \quad (11)$$

where $p(y, t_s)$ is the probability of becoming a recipient of survivor insurance with characteristics (y, t_s) , and $E(y, t_s)$ is an exogenous budget requirement.

We can analyze the problem separately for each (y, t_s) , without loss of generality, because these are exogenous characteristics.⁷ In what follows, we sometimes drop the

⁷Holding the budget constant for each (y, t_s) means that we are conceptually looking for Pareto-improving reforms. This avoids the need to specify a social welfare function across different values of

dependence on (y, t_s) to economize on notation.

The first line of (10) is utility in the “good” state, when both household members are alive. The second line is utility in the “bad” state, after the death of a household member. The constraint (11) is the government’s resource constraint.

We assume that the earnings of the future deceased, y , are subject to a contribution $\tau(y)$ financing survivor insurance, while the earnings of the future survivor, z_s , are subject to the tax-transfer system, $T(z_s)$. To simplify the problem, we assume that these decisions taken in the good state are exogenous to the design of survivor insurance, that is, to the choice of $\tau(y)$ and $B(y, t_s, t_d)$.

In the government’s resource constraint, we only consider taxes, retirement benefits, and survivor insurance benefits for survivors, and abstract from the rest of the tax-transfer system. This is because we focus on the population of survivors and, in particular, their behavioral responses to changes in the design of survivor insurance benefits.

Differentiating the objective with respect to a change in benefits B , and invoking an envelope argument for changes in z and t_r , the welfare impact of a revenue-neutral reform of survivor insurance is

$$\frac{d\mathcal{W}}{dB} = -(1-p) u'_g(c_g) \frac{d\tau}{dB} + p \left[(t_r - t_s) u'_w(c_w) \frac{\partial c_w}{\partial B} + (t_d - t_r) \beta u'_r(c_r) \frac{\partial c_r}{\partial B} \right] \quad (12)$$

$$= -(1-p) u'_g(c_g) \frac{d\tau}{dB} + p u'_w(c_w), \quad (13)$$

where the second equality follows from perfect consumption smoothing in widowhood,

$$u'_w(c_w) = \beta u'_r(c_r),$$

and from partially differentiating the survivor’s lifetime budget constraint, holding behavior fixed.

Differentiating the government budget constraint with respect to B , revenue neutrality implies

$$(1-p) \frac{d\tau}{dB} + p \left[((t_r - t_s) T'(z) - R'_z(z, t_r)) \frac{dz}{dB} + (T(z) - R'_{t_r}(z, t_r)) \frac{dt_r}{dB} - 1 \right] = 0. \quad (14)$$

(y, t_s) .

Substituting this expression into (13) yields

$$\frac{d\mathcal{W}}{dB} = -pu'_g(c_g) \left[1 - ((t_r - t_s)T'(z) - R'_z(z, t_r)) \frac{dz}{dB} - (T(z) - R'_{t_r}(z, t_r)) \frac{dt_r}{dB} \right] + pu'_w(c_w). \quad (15)$$

3.3 Sufficient Statistics Formula for Optimal Survivor Insurance

Characterizing optimal survivor insurance through $d\mathcal{W}/dB = 0$ and rearranging leads to the following optimality condition, for each (y, t_s) :

$$\frac{u'_w(c_w) - u'_g(c_g)}{u'_g(c_g)} = ((t_r - t_s)T'(z) - R'_z(z, t_r)) \left(-\frac{dz}{dB} \right) + (T(z) - R'_{t_r}(z, t_r)) \left(-\frac{dt_r}{dB} \right). \quad (16)$$

The left-hand side measures the consumption-smoothing value associated with a marginal increase in the generosity of survivor insurance. The right-hand side measures the efficiency costs stemming from reductions in earnings and in the retirement age. Reductions in earnings and in the retirement age both reduce tax revenue during working life, but this negative effect is partially offset by the associated reduction in pension benefits during retirement.

4 Labor Supply Responses to Deaths of Relatives

In this section, we present our main estimates of the effects of the death of a family member on the labor supply of individuals in the presence of survivors' pensions. We begin by describing our sample construction procedures, then we discuss our empirical strategy and present our results.

4.1 Sample Construction

Our main analysis sample comprises spouses of Public Sector Federal workers who were between 24 and 60 years old at the moment of the servant's death and had at least one job contract in the 4 years before the event. We also focus on events between the second quarter of 2003 and the end of 2014. In robustness and heterogeneity exercises, we relax these restrictions. In particular, we also use military pension recipients and recipients beyond spouses. The procedures to build the military sample are similar to the ones described below.

The main sample restrictions are designed to ensure a comparable set of individuals exposed to similar treatment throughout our analysis. The period restriction (2003–2014)

ensures that all individuals included in the event study have at least one pre-treatment observation, since name and CPF identifiers are only available in RAIS starting in 2003. We end the sample in 2014 to avoid capturing effects driven by the 2015 reform, which we analyze separately later. Our age restriction focuses on individuals of working age. Finally, we restrict the sample to individuals with at least one formal employment contract in the four years preceding the death of their relative, capturing individuals with stronger labor market attachment.⁸

To combine Pensions data with RAIS, we use a two-step procedure. First, we find all pension recipients who have an exact match on the 6-digits of CPF and date of birth with a worker in RAIS between 2003 and 2023.⁹ We then use an algorithm of text analysis to match those with the same name.¹⁰ With the sample of individuals we find in RAIS, we construct a quarterly panel of individual labor supply. We use information on admission and dismissal dates to determine whether a given job contract was active in each month. We then aggregate the contract-level data to the individual level, creating a quarterly panel. An individual is considered employed in a given quarter if they held at least one active contract during any month of that quarter. In such cases, we assign to the individual-quarter observation the average wages and hours from the active contract. If the individual held multiple contracts within the same quarter, we retain the contract with the highest wage.¹¹

4.2 Empirical Strategy

Our empirical strategy consists of an event study model using our quarterly panel of eventually treated individuals described above. The equation we estimate can be written as:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{k=-16, k \neq -1}^{16} \beta_k \cdot D_{it}^k + u_{it} \quad (17)$$

⁸Table A1 show how our sample changes according to restrictions. From the total number of beneficiaries, around 69% of them were spouses. Among those, 77% were over 55 at the moment of the event. These two criteria are responsible for the biggest reduction in the number of individuals from the total pension recipients to our analysis sample.

⁹In the case of military pensions, we only use the 6-digit CPF as we do not observe the recipients' date of birth.

¹⁰More specifically, we use the *matchit* Stata command with the *token* method and restrict those with a score above 0.7. In the case of military recipients, since we do not have their date of birth, we increase our tolerance score to 0.9.

¹¹This procedure is standard when transforming contract-level RAIS data to the individual level; see, for example, Gerard et al. (2021).

where α_i is a person fixed effects and γ_t a period fixed effect. The parameter of interest are the coefficients β_k for $k \in \{-16, -15, \dots, -2, 0, 1, \dots, 16\}$. The coefficient for the period right before the shock happened is normalized to zero (i.e. $\beta_{-1} = 0$). D_{it}^k is a dummy variable that indicates if the individual i in period t is k quarters away (or from) their relative’s death. Therefore, we interpret β_k as the difference in average labor supply in period k and the average labor supply in the period right before the death of the relative. We restrict the sample to observations in the window of $[-16, 16]$ quarters relative to the event. In all our event-studies estimates, standard errors are clustered at the individual level. We follow this standard practice as individual-specific errors are likely to be positively serially correlated.

The identifying assumption of our model is that, conditional on time and unit fixed effects, the timing of treatment is as good as random. Specifically, units that are treated later serve as valid counterfactuals for those treated earlier in the pre-treatment periods. This implies that, absent treatment, the outcome trajectories of units treated at different times would have evolved similarly. In the last part of this section, we present different robustness checks to validate our identifying assumption.

4.3 Main Results

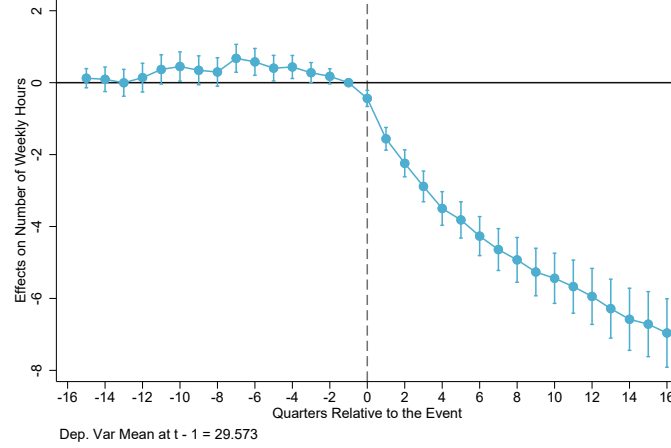
We find substantial reductions in individuals’ labor supply in response to their relative’s death in the presence of survivors’ pensions. In Figure 2, we present our event-study estimates on the number of hours worked, and both intensive and extensive margin responses. We observe that they replicate the patterns in Figure A2, now properly estimated with calendar time and individual fixed effects. In Panel (a), we show that 4 years after the event, individuals work on average 6.25 hours less per week. This represents a 22% reduction in their labor supply relative to what they worked right before the death of their relative.¹²

The labor supply reduction is mostly, but not exclusively, driven by extensive margin responses. In Panel (b), we plot the estimated effects on the number of hours worked, conditional on being employed, which we refer to as intensive margin responses. We observe a small decrease of 0.2 hours per week four years after the event—a 0.5% reduction relative to the baseline level prior to the event. In Panel (c), we plot the estimated effects on the probability of being employed (Extensive margin). We find a 16 percentage point decrease in employment probability, equivalent to a 22% reduction relative to the baseline

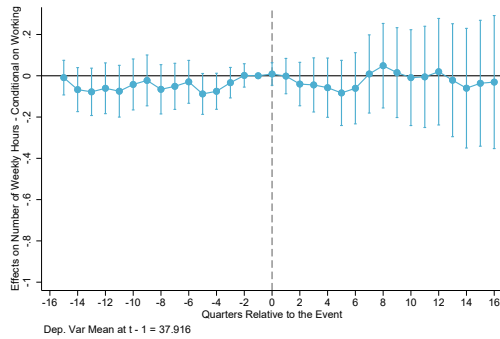
¹²In Figure A2, we plot the average hours worked and probability of employment relative to the event without relying on fixed effects. We observe that prior to the event, there is no visible pattern of changes in labor supply.

probability.¹³

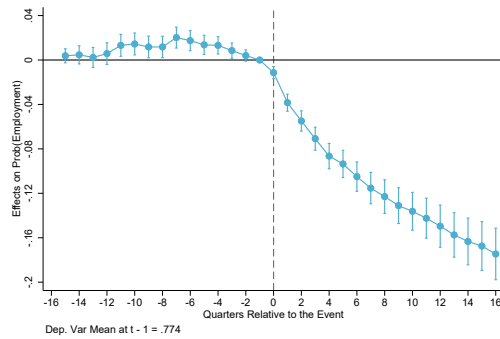
Figure 2: Effects on Labor Supply



(a) Number of Hours



(b) Intensive Margin



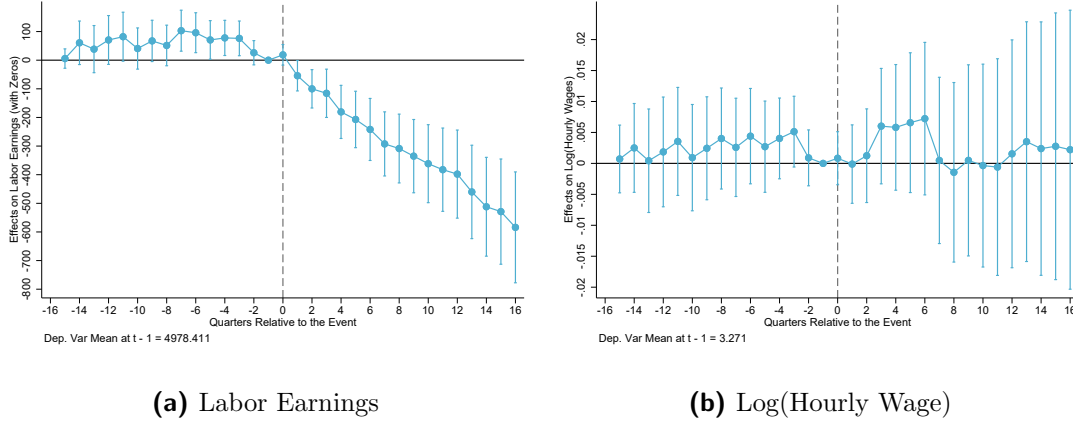
(c) Extensive Margin

This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event. Panel (b) sample is restricted to those employed in the given quarter. Standard errors are clustered at the individual level.

In Figure 3, we present our estimates of the effects on wages and labor earnings. In Panel (a), we show that individuals on average earn 583 BRL less 4 years after the death of their relative. This represents a 12% decrease in their labor earnings relative to the period before the event. In contrast, in Panel (b), we show no significant changes in log of hourly wages conditional on working.

¹³The small of intensive margin responses in hours is consistent with the rigidity of the Brazilian labor market, where over 90% of job contracts specify either 40 or 44 weekly hours.

Figure 3: Effects on Labor Earnings



This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event. In Panel (a), we input 0s when the individual is not working. In Panel (b), the sample is restricted to those employed in the given quarter. Hourly wage is calculated by dividing the monthly wage by weekly hours multiplied by 4.2. Standard errors are clustered at the individual level.

4.4 Heterogeneity and Additional Outcomes

Relation of the recipient with Former Servant: We begin by analyzing if labor supply responses are different according to the relation of the recipient with the former public sector or military worker. Figure A3 presents event-study estimates of the effects of a relative's death on weekly hours worked, separately for wives, husbands, and daughters of deceased public servants. The patterns are similar across groups, especially between wives and daughters, both of whom exhibit large and sustained declines in labor supply following the event. In both cases, the reduction in weekly hours begins immediately after the death and continues to deepen over time, reaching approximately 8 hours per week within two years. Husbands also display a decline in hours worked, though the estimates are noisier due to a smaller sample size. They, on average, also have a slightly higher baseline number of hours, thus suggesting a smaller percentage change.

The similarity in labor supply responses across dependent types—particularly between wives and daughters—suggests that the observed effects are unlikely to be driven primarily by changes in household structure or caregiving responsibilities following the death. Instead, the consistency in magnitude and timing points to the survivor pension itself as the main driver of the adjustment. While one might expect differences in labor supply trajectories if household roles were shifting substantially (e.g., spouses assuming new responsibilities or daughters altering behavior based on cohabitation), the parallel declines

across distinct demographic groups reinforce the interpretation of a direct income effect induced by the pension benefit. This strengthens the case for viewing survivor pensions as a key determinant of post-event labor market behavior.

Recipient Characteristics: Figure A4 explores heterogeneity in the labor supply response across recipient characteristics. Panel (a) shows that the reduction in weekly hours is remarkably similar for men and women, suggesting that gender does not play a major role in shaping the response to the survivor pension. In contrast, other dimensions reveal meaningful differences. The decline in hours is larger for nonwhite individuals (Panel b), those with lower levels of education (Panel c), and recipients aged 45 or older (Panel d). Finally, Panel (e) shows that private sector workers reduce their hours more than public sector workers, suggesting greater flexibility—or greater labor supply elasticity—in the private sector. Taken together, these patterns indicate that while gender responses are uniform, individuals with weaker labor market attachment or lower pre-event earnings tend to respond more strongly to the income shock provided by the pension.

No labor attachment: In our main estimates, we restrict our sample to those who had worked at least one period in the 4 years before the event. We do so to better target individuals who are in the labor force. In Figure A7 we show estimates without this restriction. In this sample, the magnitude of our findings is slightly smaller, but they also have a substantially smaller baseline. Our estimates suggest a 16% decrease in labor supply when not restricting the sample to the pool of individuals that are likely to be in the labor force.

Military vs Federal Workers: We also replicate our main estimates using the sample of *Militares* workers' relatives. We show these results in Figure A5. We observe no significant difference between our estimates, reinforcing our choice of pooling both pension schemes in our main analysis. In our main estimates, we restrict our sample to those who had worked at least one period in the 4 years before the event. We do so to better target individuals who are in the labor force. In Figure A7 we show estimates without this restriction. In this sample, the magnitude of our findings is slightly smaller, but they also have a substantially smaller baseline. Our estimates suggest a 16% decrease in labor supply when not restricting the sample to the pool of individuals that are likely to be in the labor force.

Labor Supply or Occupational Change? The focus on the formal labor market

is a potential limitation of our analysis. In RAIS data, we cannot account for other potential sources of income. We address this by using data that covers the near-universe of firm ownership in Brazil. We replicate our event-study design using two different entrepreneurial outcomes in Figure 4. In Panel (a), we show the effects of being an owner of a regular firm in period t . Regular firms are defined as all non-MEI firms. In Panel (b), we show the effects on being a registered microentrepreneur through the MEI program.¹⁴

We find that individuals are not transitioning from formal labor employment to entrepreneurship. We observe no changes when looking at the probability of owning a formal firm in Panel (a). At the same time, we find that pension recipients decrease the chances of engaging in low-value entrepreneurship, as we observe a decrease in the probability of being a MEI firm owner in Panel (b).

In particular, 97% of MEI firms have no employee (Hsu Rocha and de Farias, 2021), and are thus truly one-person firms, which demand labor supply from the entrepreneurs. Thus, the decrease in the probability of owning an active MEI firm results is consistent with the labor supply responses presented before. Individuals who receive the pensions are less likely to engage in labor activity, as their household earnings are more than compensated by the survivors' pensions they receive.

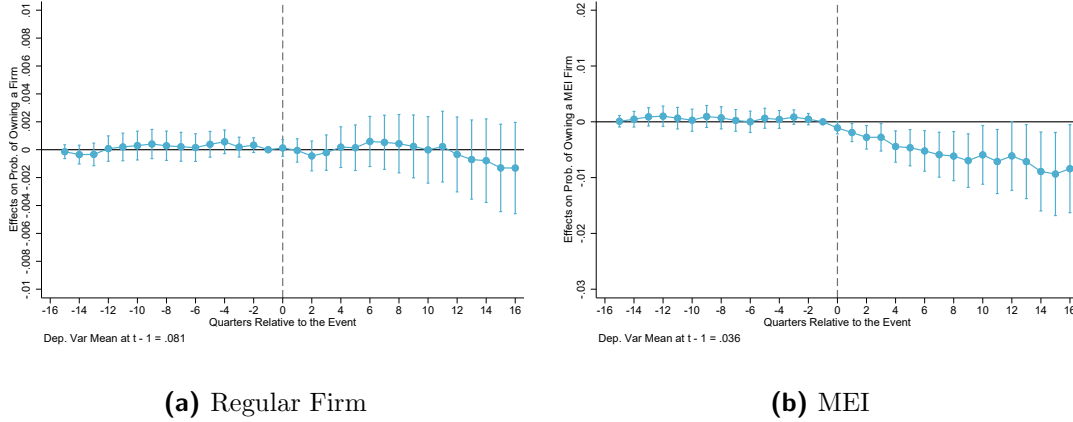
We also investigate the effects of the relative's death on the probability that the pension recipient appears in the Cadastro Único registry. Cadastro Único is a federal database that includes individuals who receive government transfers, such as the well-known conditional cash transfer program Bolsa Família. The registry covers approximately 40% of the Brazilian population, and individuals included in Cadastro Único are typically among the lower end of the income distribution or in Informal Jobs. In Figure A8, we show that following the event, pension recipients become about 20% less likely to appear in Cadastro Único, suggesting that they are less likely to be receiving other government transfers targeted to poorer households.

4.5 Robustness Checks

Our identifying assumption relies on using individuals who have not yet been exposed to the event to construct counterfactual outcomes for those who have. A growing literature has highlighted several limitations of such models in settings with staggered treatment

¹⁴MEI (or *Microempreendedor Individual*) is a specific tax system created in 2009 eligible to small microentrepreneurs with up to one worker. Hsu Rocha and de Farias (2021) studies the program and shows that such firms are substantially smaller and negatively selected relative to other formal firms in Brazil. Since the program was created in July of 2009 and was in full effect only in the middle of 2010, in Panel (b), we restrict our sample to individuals who had an event occurring after 2014, so that we have sufficient pre-event periods in our sample.

Figure 4: Effects on Entrepreneurial Activity



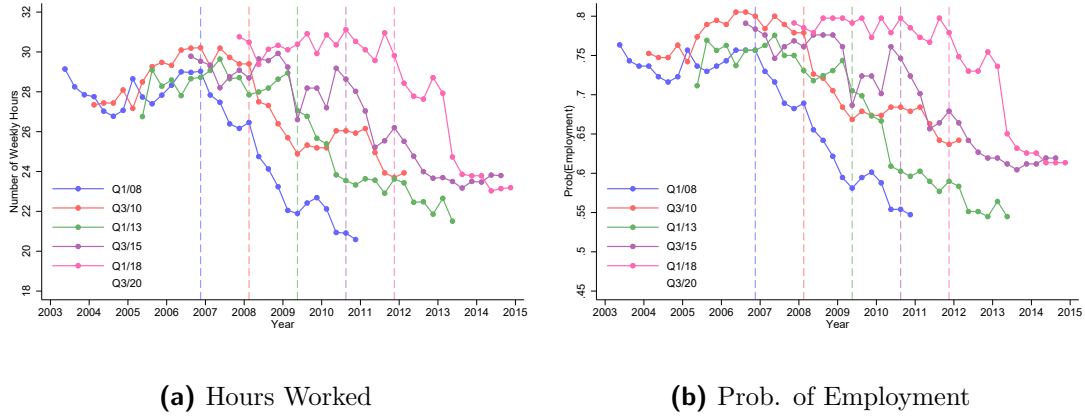
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises pension recipients between 25 and 55 years old at the time of the event. Active ownership in a given quarter t is defined by creating or joining the firm in or before period t , and the firm still being active in period t . In Panel (b), we restrict the sample to recipients who had events after 2014, as the MEI program started in July 2009.

timing and heterogeneous treatment effects (De Chaisemartin and d’Haultfoeulle, 2020; Goodman-Bacon, 2021; Callaway and Sant’Anna, 2021; Sun and Abraham, 2021). In particular, standard TWFE estimators can produce biased estimates due to contamination from already-treated units and inappropriate weighting across groups. To address these concerns, we estimate treatment effects using the improved doubly robust difference-in-differences estimator proposed in Sant’Anna and Zhao (2020), implemented via inverse probability tilting and weighted least squares. Figure A9 shows the treatment effects on our main outcomes. We observe that the estimates are extremely similar to those in our main results, indicating that issues pointed out in the recent TWFE literature are not driving our findings.

In addition, in Figure 5 we plot our labor supply measures across calendar time by event cohort. Each of the connected series in the Figure shows the average labor supply of individuals who had events in different periods. For visualization purposes, we restrict to five different event cohorts with a 5-quarter distance between them. We observe that the labor supply of individuals is remarkably similar prior to the event for the different cohorts, and follows the reduction pattern after the event happens. We take this as reassuring evidence of our event-study model.

Another empirical discussion in our setting concerns the use of outcomes with zeros. Chen and Roth (2024) discusses the applications of log linearizing regressions models with outcomes that are weakly positive and suggests that Poisson transformations are a

Figure 5: Robustness - Raw Data by Event Cohort



This Figure plots the average hours worked and probability of being employed by event cohort. Each vertical line corresponds to the event time for the respective event-cohort. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event.

potentially powerful tool to recover relative treatment effects. In our main exposition, we divide simple OLS estimates of weakly positive outcomes given that event-studies relying on staggered adoption might pose additional identifying assumptions in the case of Poisson regressions (Moreau-Kastler, 2025). However, as a robustness check, we also show our main estimates using Poisson models in Figure A10. Reassuringly, our estimates are remarkably similar to the ones we find in the simple OLS.

5 Effects of Duration of Benefits on Labor Supply Responses

In this section, we investigate how the duration of benefits affects the labor supply responses of pension recipients. We study a reform to survivor pensions in Brazil's public-sector pension system (*Regime Próprio de Previdência Social* – RPPS) that reduced the expected duration of benefits for surviving spouses of deceased federal civil servants. The reform was initially introduced through *Medida Provisória* No. 664, issued on December 30, 2014, and became effective on March 1, 2015.¹⁵

Prior to the reform, surviving spouses of federal public servants were generally entitled

¹⁵*Medidas Provisórias* are executive decrees with immediate legal force in Brazil and do not require prior congressional approval to become effective. They must subsequently be approved by Congress to become permanent law. In this case, the measure was later converted into Law No. 13,135 on June 17, 2015. While the reform was publicly announced in December 2014, we focus on March 2015 as the relevant implementation date, when the new rules began applying to newly granted survivor pensions.

Table 1: Changes in Duration of Pensions

	(1)	(2)
	Before Reform	After Reform
Age of Surviving Spouse at Death		
< 21	Lifetime	3 years
21-26	Lifetime	6 years
27 - 29	Lifetime	10 years
30 - 40	Lifetime	15 years
41-43	Lifetime	20 years
≥ 44	Lifetime	Lifetime

This table summarizes the changes in the duration of survivors' pensions to spouses of public servants generated by the 2015 law changes.

to lifetime survivor pensions, regardless of age. The reform introduced minimum marriage and contribution requirements and, most importantly for our analysis, replaced lifetime benefits for younger spouses with an age-dependent schedule that substantially reduced benefit duration.

In particular, only spouses aged 44 or older at the time of the servant's death retained the right to a lifetime pension. For younger spouses, the benefit duration became progressively shorter depending on age, ranging from 3 to 20 years. Table 1 summarizes the differential exposure to the reform. ¹⁶

5.1 Empirical Strategy

A central challenge in our empirical setting is that labor supply responses evolve along two distinct time dimensions. First, treatment exposure is determined in calendar time: spouses younger than 44 whose partners died after March 2015 were exposed to shorter pension durations. Second, labor market outcomes are measured in event time, as we observe labor supply at multiple horizons before and after widowhood. This creates a setting in which outcomes vary both relative to the timing of the reform and relative to the timing of the death event.

Because our objective is to trace labor supply responses at specific horizons relative to widowhood, we organize the data in wide format, with each outcome corresponding to a

¹⁶In addition to changes in duration, the reform introduced a minimum requirement of 18 months of contributions and at least two years of marriage or stable union, unless the death resulted from an accident. These changes aimed to reduce long-term pension liabilities and align public sector rules more closely with those in the general social security regime (RGPS).

distinct period before or after the death event. This structure allows us to estimate separate regressions for each horizon and directly compare labor market responses over time. More importantly, it allows us to cleanly separate variation driven by the timing of the reform from variation driven by the timing of widowhood, which are distinct dimensions in our setting.

Our baseline specification is a difference-in-differences design estimated at the spouse level. Let D_i denote an indicator equal to one if spouse i was younger than 44 at the time of their partner's death, and let $Post_{t(i)}$ denote an indicator equal to one if the death occurred after the implementation of the reform in March 2015. We estimate:

$$Y_i^k = \alpha + \delta_{t(i)} + \gamma D_i + \beta (Post_{t(i)} \times D_i) + \Gamma X_i + \varepsilon_i \quad (18)$$

where Y_i^k denotes outcome k periods relative to the death event, $\delta_{t(i)}$ are calendar-time fixed effects based on the timing of the spouse's death, and X_i includes demographic controls such as age, gender, race, and education.

The coefficient of interest, β , captures whether spouses exposed to shorter pension durations after the reform exhibit differential labor market outcomes relative to older spouses whose lifetime eligibility remained unchanged.

To assess the validity of the identifying assumption, we estimate a more flexible specification that allows treatment effects to vary across pre-reform periods:

$$\begin{aligned} Y_i^k = & \alpha + \delta_{t(i)} + \gamma D_i \\ & + \beta_{2004-2006} \cdot \mathbb{I}[t(i) \in [2004, 2006]] \cdot D_i \\ & + \beta_{2007-2009} \cdot \mathbb{I}[t(i) \in [2007, 2009]] \cdot D_i \\ & + \beta_{2010-2012} \cdot \mathbb{I}[t(i) \in [2010, 2012]] \cdot D_i \\ & + \beta_{post} \cdot Post_{t(i)} \cdot D_i \\ & + \Gamma X_i + \varepsilon_i \end{aligned} \quad (19)$$

The omitted category corresponds to deaths occurring between 2013 and February 2015, immediately before the reform.

Measuring Labor Supply: We measure the effects on labor supply using two different strategies. First, we estimate specifications using outcomes in levels, where Y_i^k denotes the labor market outcome of spouse i measured k periods relative to the death event. These outcomes include measures such as employment status, earnings, and labor force participation observed at different horizons after widowhood. We also estimate specifications using long differences, where the dependent variable is defined as $Y_i^k - Y_i^{-1}$,

where Y_i^{-1} denotes the same labor market outcome measured in the period immediately preceding widowhood. This outcome captures the change in labor supply between the pre-widowhood period and period k after the death event.

The long-difference specification offers an important empirical advantage in our setting. Because our data are organized in wide format, with separate outcomes measured at different horizons relative to widowhood, we cannot estimate a standard panel specification with individual fixed effects while simultaneously tracing labor supply responses over time. By subtracting pre-widowhood labor supply from post-widowhood outcomes, the long-difference specification effectively removes time-invariant individual determinants of labor supply, such as persistent differences in labor market attachment, preferences for work, and other fixed characteristics that may differ across younger and older spouses.

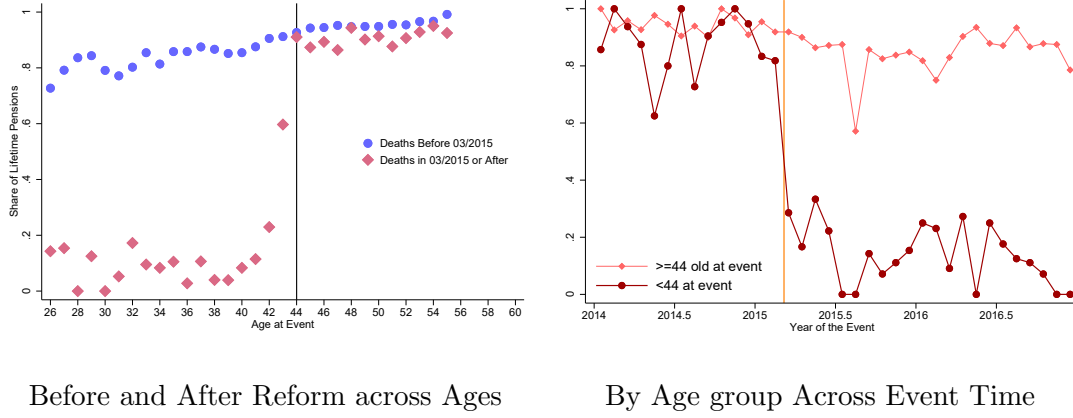
Under the assumption that pre-widowhood labor supply is not differentially affected by the reform across treatment groups, the long-difference specification recovers the causal effect of shorter pension duration on post-widowhood labor supply while netting out baseline differences across individuals. This is particularly useful in our context because spouses above and below the age threshold may differ systematically in their underlying labor market participation even in the absence of the reform. At the same time, if labor supply adjusts prior to widowhood due to anticipation of household shocks or changes in expectations, the long-difference estimates may capture both pre- and post-widowhood adjustments. For this reason, we view the levels and long-difference specifications as complementary.

Analysis Sample: To build our analysis sample, we restrict the sample from our previous exercises to spouses of former federal public sector workers. We also exclude recipients after November 2019, as they are affected by the second reform discussed below. Table [A2](#) shows summary statistics of our sample divided by the four main groups of the difference in differences. There are small differences within both treated and control groups across the samples who had events before and after 2016. To account for these differences, we control for gender, education and race in our estimations, in addition to the fixed effects by age at pension start and quarter of pension start.

5.2 Results

We begin by showing descriptive statistics on the *first stage* of the reform, specifically how the share of lifetime pensions changed according to the age of the surviving spouses in our data. Figure [6](#) presents the share of lifetime pensions by spouse’s age at the time of the public servant’s death, distinguishing between pensions initiated before and after

Figure 6: Share of Lifetime Pensions by Spouse's Age at Death



This Figure plots the share of Lifetime Pensions recipients in our data by age of recipient at the pension start date. Blue circles show pensions started before 2016, and Red diamonds show pensions started in 2016 or after. The sample comprises all spouses from former Executive Federal workers between 18 and 60 years old.

the reform's implementation in 2016. Despite compliance not being perfect, we observe a sharp reduction in the share of lifetime pensions precisely at the reform cutoff age of 44. Specifically, spouses younger than 44 who started receiving pensions after the reform experienced a marked drop from nearly universal lifetime coverage to close to zero, aligning closely with the policy's intended impact.

We find that a smaller duration of lifetime pensions increases the labor supply of recipients after the event. Table 3 presents our main estimates. Column (1) provides a first-stage validation of the institutional change. The coefficient on *Younger 44 x Post Reform* indicates that spouses younger than 44 who became widowed after the reform were 71 percentage points less likely to receive a lifetime pension relative to older spouses, consistent with the sharp reduction in benefit duration introduced by the reform. Reassuringly, the coefficients for the pre-reform interactions are small and statistically insignificant, providing support for the identifying assumption that younger and older spouses followed similar trends prior to the policy change.

Columns (2) through (7) examine labor supply responses. In the levels specifications, exposure to shorter pension duration increases weekly hours worked by approximately 4.9 hours, raises the probability of employment by 11.4 percentage points, and increases monthly labor earnings by roughly R\$1,369. Relative to the mean outcomes of the treated group, these effects are economically meaningful. The long-difference specifications yield smaller but still statistically significant estimates: weekly hours increase by approximately

Table 2: Effects of Pensions' Duration on Labor Supply Two Years After Event

	(1) Prob(Lifetime Pension)	(2) Weekly Hours	(3) Weekly Hours	(4) Prob(Employment)	(5) Prob(Employment)	(6) Labor Earnings	(7) Labor Earnings
		Levels	Δ	Levels	Δ	Levels	Δ
Younger 44 x [2004,2006]	-0.0173 (0.02260)	0.8400 (1.38633)	-0.4147 (1.43964)	0.0223 (0.03395)	-0.0055 (0.03391)	481.2081 (410.37284)	88.1297 (248.51877)
Younger 44 x [2007,2009]	-0.0096 (0.02224)	0.0300 (1.37008)	-0.4378 (1.41670)	0.0006 (0.03371)	-0.0183 (0.03355)	140.1076 (418.84761)	-337.5825 (260.39436)
Younger 44 x [2010,2012]	-0.0106 (0.02321)	1.1475 (1.38601)	0.1372 (1.44413)	0.0261 (0.03408)	-0.0014 (0.03440)	200.7938 (422.39397)	-40.8528 (255.25248)
Younger 44 x Post Reform	-0.7099*** (0.02232)	4.9256*** (1.35523)	3.0577** (1.39174)	0.1139*** (0.03377)	0.0609* (0.03348)	1368.8424*** (425.33529)	537.8700** (266.11185)
Mean of Outcome (Treated)	.9	18.8	18.8	.49	.49	3467.04	3467.04
Mean of Δ Outcome (Treated)			-5.74		-13		-398.49
Observations	9499	9499	9499	9499	9499	9499	9499

This Table shows the estimates of β of equation 19 using labor supply 2 years after the event as the outcome. We include education, gender, and race dummies as controls, in addition to Fixed Effects of quarter of benefit start, and Age at Start. The sample comprises all spouses from former Executive Federal workers who started receiving benefits between 2003 and 2018. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

3 hours, employment rises by 6.1 percentage points, and labor earnings increase by roughly R\$538. The smaller magnitudes in the long-difference specifications suggest that part of the levels estimates may reflect persistent differences in labor market attachment across younger and older spouses. Nevertheless, both specifications consistently indicate that shorter survivor pension duration leads surviving spouses to increase labor supply following widowhood.

Table A4 examines labor market outcomes in the year prior to widowhood. An important empirical question in our setting is whether spouses adjust labor supply in anticipation of the death event, particularly if households can foresee deteriorating health conditions or anticipate future changes in pension income. We find no evidence of such anticipatory responses. The coefficients on *Younger 44 x Post Reform* are small and statistically insignificant across weekly hours, employment, and labor earnings in both the levels and long-difference specifications. The pre-reform interaction coefficients are also generally close to zero, providing additional evidence against differential pre-trends between younger and older spouses for events prior to the reform. Taken together, these results suggest that the labor supply effects documented after widowhood primarily reflect post-shock adjustments rather than changes occurring prior to the death event.

6 Effects of Value of Benefits on Labor Supply Responses

In this section, we analyze how the generosity of survivor pensions affects labor supply. We exploit a reform enacted through Constitutional Amendment No. 103/2019, which substantially reduced the value of survivor pensions for new beneficiaries in Brazil's public

sector pension system (RPPS).

Prior to the reform, surviving dependents in RPPS typically received close to full income replacement. Survivor pensions paid 100 percent of the deceased worker’s retirement benefit up to the social security ceiling and 70 percent of any amount above that threshold. In practice, this implied near-full replacement for many beneficiaries, particularly among public servants with relatively high earnings histories.

Beginning in November 2019, the reform introduced a less generous replacement rate for new beneficiaries. Survivor pensions became equal to a base payment of 50 percent of the deceased worker’s retirement benefit, plus an additional 10 percentage points for each eligible dependent, capped at 100 percent. For example, a surviving spouse with no children would receive only 60 percent of the original benefit, while households with multiple dependents would experience smaller reductions. Importantly, the reform altered the value of benefits but did not modify the duration rules introduced in 2015.

6.1 Empirical Strategy

To identify causal effects of the reform, we exploit the fact that military pensions were exempt from the reform and maintained substantially more generous rules during this period. We therefore use survivors of military personnel as a comparison group for survivors of public servants affected by the reform.

We proceed in a very similar fashion to our empirical strategy for the 2015 reform. Essentially, the dimensionality aspects of the problem are identical. The only difference is in the sample and how to define the treatment and control groups. This implies we will also focus on two sets of outcomes: levels and long differences, for the same reasons that applied before in our 2015 reform exercise.

We estimate a difference-in-differences model comparing spouses exposed to the reform in the RPPS system to spouses of military workers, before and after November 2019.

Let D_i denote an indicator equal to one if individual i is the surviving spouse of a public servant covered by RPPS, and zero if the individual is the surviving spouse of a military worker. Let $Post_{t(i)}$ denote an indicator equal to one if the death event occurred after the implementation of the reform in November 2019. We estimate:

$$Y_i^k = \alpha + \delta_{t(i)} + \gamma D_i + \beta (Post_{t(i)} \times D_i) + \Gamma X_i + \varepsilon_i \quad (20)$$

where Y_i^k represents labor market outcomes measured k periods relative to the death event, $\delta_{t(i)}$ captures calendar-time fixed effects based on the timing of widowhood, and X_i includes demographic controls such as age, gender, race, and education.

The coefficient of interest, β , measures whether surviving spouses exposed to the

reduction in pension generosity experienced differential labor supply responses relative to the military comparison group.

We estimate a specification that allows for differential pre-trends between the two groups. Specifically, we interact the RPPS indicator with a pre-reform period indicator for deaths occurring between January 2016 and December 2018:

$$Y_i^k = \alpha + \delta_{t(i)} + \gamma D_i + \beta_{pre} (Pre_{t(i)} \times D_i) + \beta_{post} (Post_{t(i)} \times D_i) + \Gamma X_i + \varepsilon_i \quad (21)$$

where $Pre_{t(i)}$ equals one for deaths occurring between January 2016 and December 2018.

Analysis Sample: Our analysis sample comprises all spouses of Federal and Military workers who received benefits between January 2016 and December 2021. We exclude those before 2016 as Federal workers would be on different pension duration schemes, given the 2015 reform. We also restrict our sample in December 2021 because of RAIS data availability, as we want to observe their labor supply up to two years after the event (currently, 2023 is the last year available).

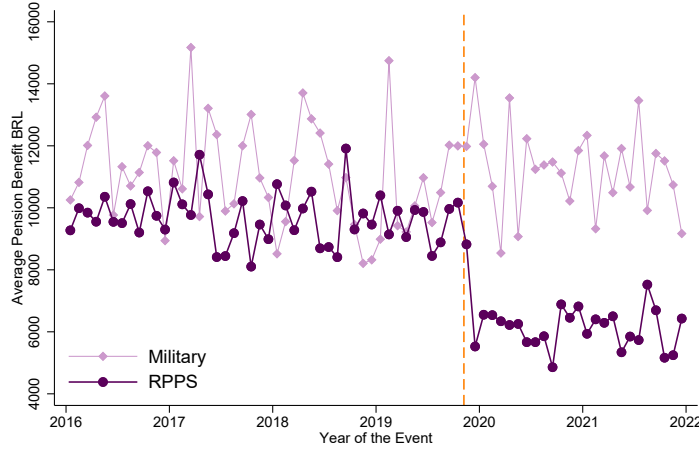
6.2 Results

We begin by showing descriptive values on the *first-stage* of the reform. In Figure 7, we show the average value of pension benefits for spouses of Military and Federal workers between 2016 and 2021. We observe that federal workers' surviving spouses face a sudden and large drop in benefit values in November 2019, going from around 10 thousand BRL to slightly above 6 thousand. Meanwhile, we see no differential pattern during the selected time frame for surviving spouses of Military workers.

Table ?? presents the effects of the 2019 reform on survivor pension generosity and labor supply outcomes. Column (1) confirms that the reform substantially reduced the benefit values: treated spouses received 33% less benefits after the reform. The estimates for the intermediate pre-reform period ($Treated \times [2016-2018]$) are small and statistically insignificant, suggesting no differential trends prior to the policy change.

Columns (2) through (7) examine labor supply responses. In contrast to the 2015 reform, the effects in levels are smaller and imprecisely estimated. However, the long-difference specifications show meaningful labor supply responses: treated spouses increase weekly hours by approximately 3.8 hours, employment by 8.1 percentage points, and labor earnings by roughly R\$977 following widowhood. Relative to the average decline in labor

Figure 7: Average value of Pensions Benefits by Date of Servant's Death RPPS vs Military



This Figure plots the average Gross Pension Benefit by the month in which the benefit started separately for spouses of Federal (RPPS) and Military workers. The orange dashed line marks November 2019, the moment in which the reform was implemented. Nominal values adjusted by inflation to 2023 BRL.

supply observed among treated spouses after widowhood, these estimates imply that the reduction in pension generosity partially offsets the decline in labor market activity following the death event. Taken together, these results provide additional evidence that lower survivor benefits increase labor supply among surviving spouses.

In Table A5 we show labor market outcomes one year prior to widowhood for the 2019 reform. As in the previous setting, an important empirical question is whether spouses adjust labor supply in anticipation of the death event or anticipated changes in pension generosity. We find no evidence of such anticipatory responses. The coefficients on *Treated*

Table 3: Effects of Pensions' Duration on Labor Supply Two Years After Event

	Log(Benefits)	Weekly Hours		Prob(Employment)		Labor Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Levels	Δ	Levels	Δ	Levels	Δ
Treated x [2016,2018]	-0.0712 (0.06069)	-0.7381 (1.50700)	1.5389 (1.44956)	-0.0165 (0.03905)	0.0140 (0.03649)	-208.7961 (427.14440)	400.7298 (341.35247)
Treated x Post Reform	-0.3331*** (0.06449)	1.1915 (1.56195)	3.8285** (1.49966)	0.0466 (0.03970)	0.0812** (0.03747)	136.0462 (480.67897)	976.5203** (430.45849)
Mean of Outcome (Treated)	9.44	18.93	18.93	.51	.51	4106.4	4106.4
Mean of Δ Outcome (Treated)			-7.26		-.17		-1572.83
Observations	6612	6663	6663	6663	6663	6663	6663

This Table shows the estimates of β of equation 21 using labor supply 2 years after the event as the outcome. We include education, gender, and race dummies as controls, in addition to Fixed Effects of quarter of benefit start, and Age at Start. The sample comprises all spouses of former Executivo Federal and Military workers who started receiving benefits between 2016 and 2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

x *Post Reform* are small and statistically insignificant across weekly hours, employment, and labor earnings in both the levels and long-difference specifications, and the estimates for the intermediate pre-reform period (*Treated* x [2016–2018]) are also generally close to zero. These findings reinforce the interpretation of our previous estimates by showing that the labor supply responses emerge only after widowhood, rather than reflecting pre-existing trends or anticipatory adjustments prior to the death event.

7 Discussion

In this section, we refer back to our conceptual framework and discuss the implications of our reduced form estimates.

7.1 Quantifying the Effect: Labor Supply Elasticities

From our conceptual framework, we derived a sufficient statistics formula for the optimal survivors’ insurance, re-stated below. It depended on the left-hand side of the change in marginal utility from the good state to the widowhood state, and on the right-hand side of the equation on the externalities on the government budget caused by individuals’ behavioral response.

$$\frac{u'_w(c_w) - u'_g(c_g)}{u'_g(c_g)} = ((t_r - t_s)T'(z) - R'_z(z, t_r)) \left(-\frac{dz}{dB} \right) + (T(z) - R'_{t_r}(z, t_r)) \left(-\frac{dt_r}{dB} \right). \quad (22)$$

To interpret our findings in light of our sufficient statistics formula, we need a few steps. First, we need to convert the changes from the reforms into changes in lifetime benefits dB . This is somewhat trivial to the 2019 reform, but not for the 2015 reform on benefit duration. Having recovered our estimates for dB , we can then translate this into labor supply elasticities, needed for recovering the externalities on the government budget. Lastly, we turn to the left-hand side and discuss how to link our labor supply responses to the spouse’s death with the changes in marginal utility from the good to the widowhood state.

Recovering dB from 2015: To translate this policy shift into a money metric for the change in benefits, we use official life expectancy data from the Brazilian Institute of Geography and Statistics (IBGE). The average individual in our sample of treated individuals has a life expectancy of 45.2 additional years after the death of their spouse.¹⁷

¹⁷We present in Table A3 the values in 2015, for the age groups of interest around the reform. We

Thus, the average change is from 45.2 to 15 years of receiving benefits.

To recover dB we can thus write:

$$dB = \frac{\sum_{t=0}^{t=45} \beta^t b - \sum_{t=0}^{t=15} \beta^t b}{\sum_{t=0}^{t=45} \beta^t b}$$

where b corresponds to the annual benefit received and β the intertemporal discount rate. Under a standard 0.96 intertemporal discount rate, the 2015 reform represents a 43.4% reduction in total benefits.

Recovering dB from 2019: Recovering the change in benefits caused by the 2019 reform is more straightforward and involves fewer assumptions. As the reform basically changes the value of benefits for recipients in all periods we can simply write:

$$dB = \frac{\sum_t \beta^t b - \sum_t \beta^t (1 - 0.3331) \cdot b}{\sum_t \beta^t b}$$

where the intertemporal discount cancels out, and 0.3331 is recovered from our estimates of the effects of the policy on total lifetime benefits.

Recovering Elasticities: We approximate labor supply elasticities using our estimates of the changes in labor supply from the reform. We can write the elasticity we recover ε as:

$$\varepsilon = \frac{\partial y}{y} \cdot \frac{\text{Benefits}}{\partial \text{Benefits}} = \frac{\Delta y}{\bar{y}} \cdot \frac{\text{Benefits}}{\Delta \text{Benefits}} \quad (23)$$

We report our elasticity estimates in Table 4. To recover the elasticities with respect to the 2015 reform, we divide our reduced form estimates on the changes in labor supply outcomes by the first stage on the duration of the reforms. We then use the baseline dependent variable mean to recover the percent change in labor supply and calculate the elasticities. A similar procedure is made with respect to the 2019 reform, with the key exception that our first stage is already implicit in the changes of benefits.

Table 4: Elasticities of Labor Supply with Respect to Lifetime Benefits

	dB	Hours	Prob(Employment)	Earnings
2015 Reform ($\beta = 0.96$)	43.4%	- 0.5279	- 0.4034	- 0.5031
2019 Reform	33.3%	- 0.6128	- 0.4825	- 0.7206

combine the share of our sample by gender and the average age to calculate the average life expectancy in our sample of treated individuals.

Our elasticity estimates are broadly consistent with recent work documenting sizable effects from survivor pensions, though our setting and margin of variation differ in important ways. [Giupponi \(2019\)](#) studies a reform in Italy that reduced the per-period amount of survivor pensions and finds an elasticity $-.8$, closely aligned to our estimate for the 2019 reform, which also leverages a per-period change in benefit. [Coyne et al. \(2024\)](#) simulate labor supply responses to the loss of U.S. Social Security survivors' benefits and estimate that households substantially value and adjust behavior around these benefits, especially among lower earners.

Going Back to Sufficient Statistics Formula

We now return to the sufficient-statistics empirical estimates to provide a preliminary welfare interpretation of survivor pensions. Recall that the optimality condition highlights a trade-off between the value of insuring across states and the fiscal costs generated by behavioral responses.

We begin with the left-hand side of the formula. Recall that for a given benefit amount, an increase in the marginal utility of consumption, approximated by an increase in labor supply, represents that individuals are willing to pay to bring resources from the good to the widowhood estimates. On the other hand, a reduction in the marginal utility of consumption, approximated by a reduction in labor supply, indicates that individuals are over-insured and they are willing to pay to bring some of the benefits in the widowhood state to the good state. Given that survivor insurance is so high in this context, our event-study estimates provide evidence that widowhood reduces labor supply for those individuals receiving the benefit. Therefore, even in the absence of fiscal externalities, our estimates suggest that survivorship insurance should be reduced.

However, this does not mean that insurance should be zero. [Fadlon and Nielsen \(2021\)](#) shows that, absent survivorship pension, widows considerably increase their labor supply. While we cannot directly estimate our event-study for those who do not receive pensions, cross-sectional analysis using the Brazilian household survey also shows that widows with no benefits have a larger labor supply than similar non-widows, whereas widows receiving benefits have a smaller labor supply. Therefore, the optimal insurance is neither zero nor the amount that fully insures individuals, because of fiscal externalities.

Then, we turn to the fiscal externality term on the right-hand side of the formula. For now, we focus exclusively on labor supply responses during working life and abstract from retirement responses.¹⁸

To quantify the fiscal externality from labor supply reductions, we combine our esti-

¹⁸We are working on incorporating this margin.

mates with Brazil’s tax schedule on formal labor earnings. Formal employment is subject to a 20 percent employer payroll tax, employee social security contributions that range from roughly 7 to 14 percent, and income taxation that can reach 27.5 percent for higher earners. Using a conservative effective marginal tax rate of 27 percent, our estimates imply fiscal costs equivalent to approximately 18.9 percent of survivor benefits through reduced labor supply alone.

Taken together, these calculations suggest the optimal amount of survivorship insurance should imply an increase in labor supply mapping to an increase in the marginal utility of consumption equivalent to the fiscal externality. It is worth flagging some simplifications in our analysis that may imply different welfare consequences. First, our measure of the left-hand side captures only the immediate consumption-smoothing value of insurance and does not fully account for broader welfare gains associated with reducing financial distress after widowhood. Second, our current calculation abstracts from retirement responses, which could either amplify or partially offset fiscal costs depending on how survivor benefits interact with future pension claiming. More broadly, these results suggest that while generous survivor pensions provide valuable insurance against a severe household shock, there is meaningful scope to redesign benefit generosity along dimensions such as duration and replacement rates to better balance insurance provision and labor market distortions.

8 Conclusion and Future Work

This paper studies the labor supply responses of individuals to the death of family members in the context of generous survivor pension schemes in Brazil. Using administrative data from survivor pension recipients and matched employer-employee records, we document substantial reductions in labor supply and earnings following the relative’s death.

Overall, our findings underscore the significant impact of survivor pensions on labor supply decisions and suggest potential fiscal and efficiency gains from carefully designed reforms to pension generosity and duration. These insights are crucial for policymakers aiming to balance social insurance objectives with labor market incentives.

Bibliography

- Callaway, Brantly and Pedro HC Sant’Anna**, “Difference-in-differences with multiple time periods,” *Journal of econometrics*, 2021, 225 (2), 200–230.
- Chaisemartin, Clément De and Xavier d’Haultfoeuille**, “Two-way fixed effects estimators with heterogeneous treatment effects,” *American Economic Review*, 2020, 110 (9), 2964–96.
- Chen, Jiafeng and Jonathan Roth**, “Logs with zeros? Some problems and solutions,” *The Quarterly Journal of Economics*, 2024, 139 (2), 891–936.
- Coyne, David, Itzik Fadlon, Shanthi P Ramnath, and Patricia K Tong**, “Household labor supply and the value of social security survivors benefits,” *American Economic Review*, 2024, 114 (5), 1248–1280.
- Fadlon, Itzik and Torben Heien Nielsen**, “Family Labor Supply Responses to Severe Health Shocks: Evidence from Danish Administrative Records,” *American Economic Journal: Applied Economics*, 2021.
- Gerard, François, Lorenzo Lagos, Edson Severnini, and David Card**, “Assortative matching or exclusionary hiring? The impact of employment and pay policies on racial wage differences in Brazil,” *American Economic Review*, 2021, 111 (10), 3418–3457.
- Giupponi, Giulia**, “When income effects are large: labor supply responses and the value of welfare transfers,” 2019.
- Goodman-Bacon, Andrew**, “Difference-in-differences with variation in treatment timing,” *Journal of econometrics*, 2021, 225 (2), 254–277.
- Moreau-Kastler, Ninon**, “Proportional Treatment Effects in Staggered Settings: An Approach for Poisson Pseudo-Maximum Likelihood,” *Available at SSRN 5247936*, 2025.
- Rabaté, Simon and Julie Tréguier**, “Labour supply and survivor insurance in the Netherlands,” *Labour Economics*, 2024, 88, 102527.
- Rocha, Roberto Hsu and Alison de Farias**, “Formality cost, registration and development of microentrepreneurs: Evidence from brazil,” *Registration and Development of Microentrepreneurs: Evidence From Brazil (November 10, 2021)*, 2021.

Sant’Anna, Pedro HC and Jun Zhao, “Doubly robust difference-in-differences estimators,” *Journal of econometrics*, 2020, *219* (1), 101–122.

Sun, Liyang and Sarah Abraham, “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects,” *Journal of econometrics*, 2021, *225* (2), 175–199.

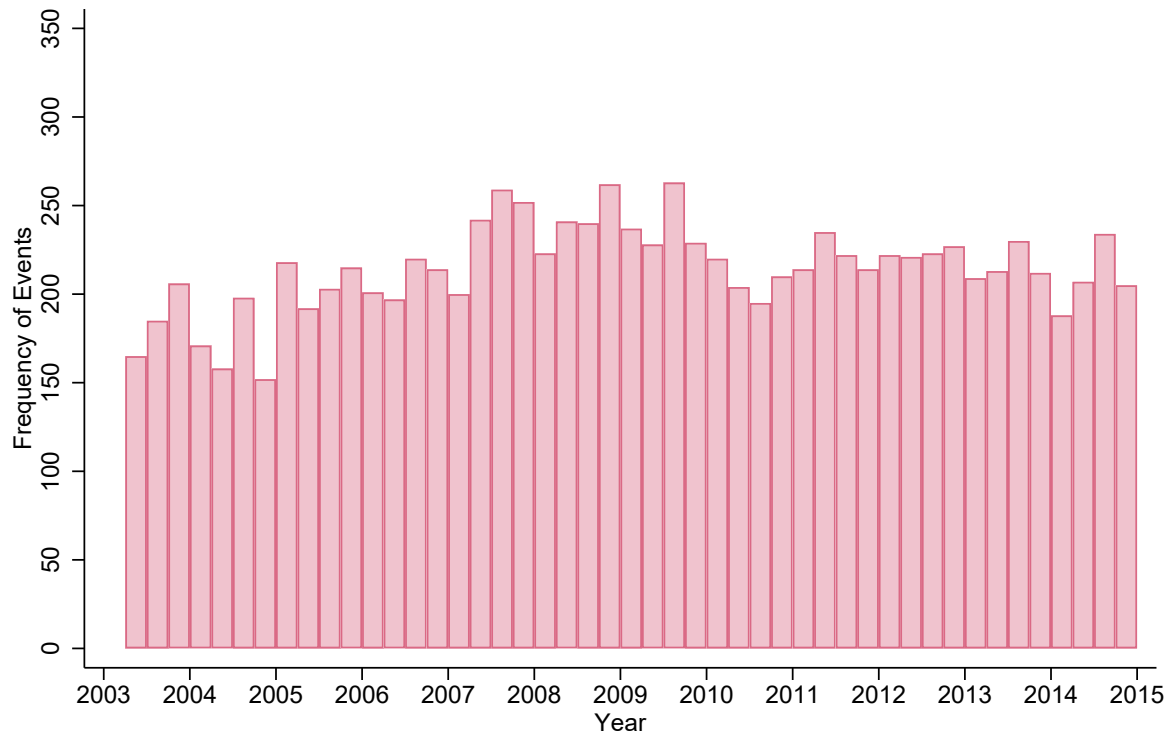
9 Additional Results

Table A1: Building Main Analysis Sample

	(1)	(2)	(3)	(4)	(5)
	All	Spouses	24-55	Age + Spouses	+ Labor Force
Monthly pension benefit (BRL)	6351.462	7034.539	6171.171	6413.551	7174.387
Spouse beneficiary	0.687	1.000	0.632	1.000	1.000
Lifetime pension benefit	0.738	0.998	0.735	0.997	0.996
Age at pension start	54.468	64.757	45.131	46.321	45.944
Age > 55 at pension start	0.567	0.766	0.000	0.000	0.000
Age < 24 at pension start	0.180	0.002	0.000	0.000	0.000
Female beneficiary	0.195	0.107	0.168	0.099	0.167
Worked in quarter before event	0.049	0.053	0.193	0.228	0.774
Observations	154794	106331	39058	24675	7252

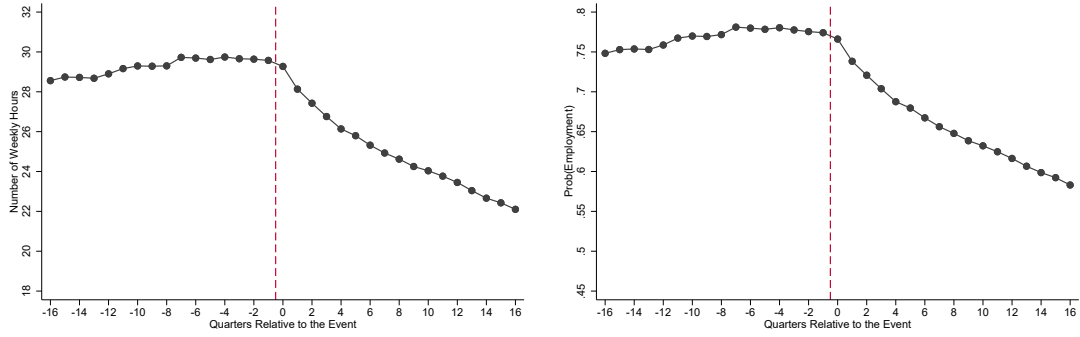
This table presents summary statistics of our main analysis sample. Labor market information corresponds to the quarter before the event. Benefits and earnings values are adjusted to Dec. 2023 Brazilian Reais.

Figure A1: Histogram of Events in the Analysis Sample



This Figure shows a histogram of Event Frequency by Quarter. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event.

Figure A2: Average Labor Supply Relative to Event

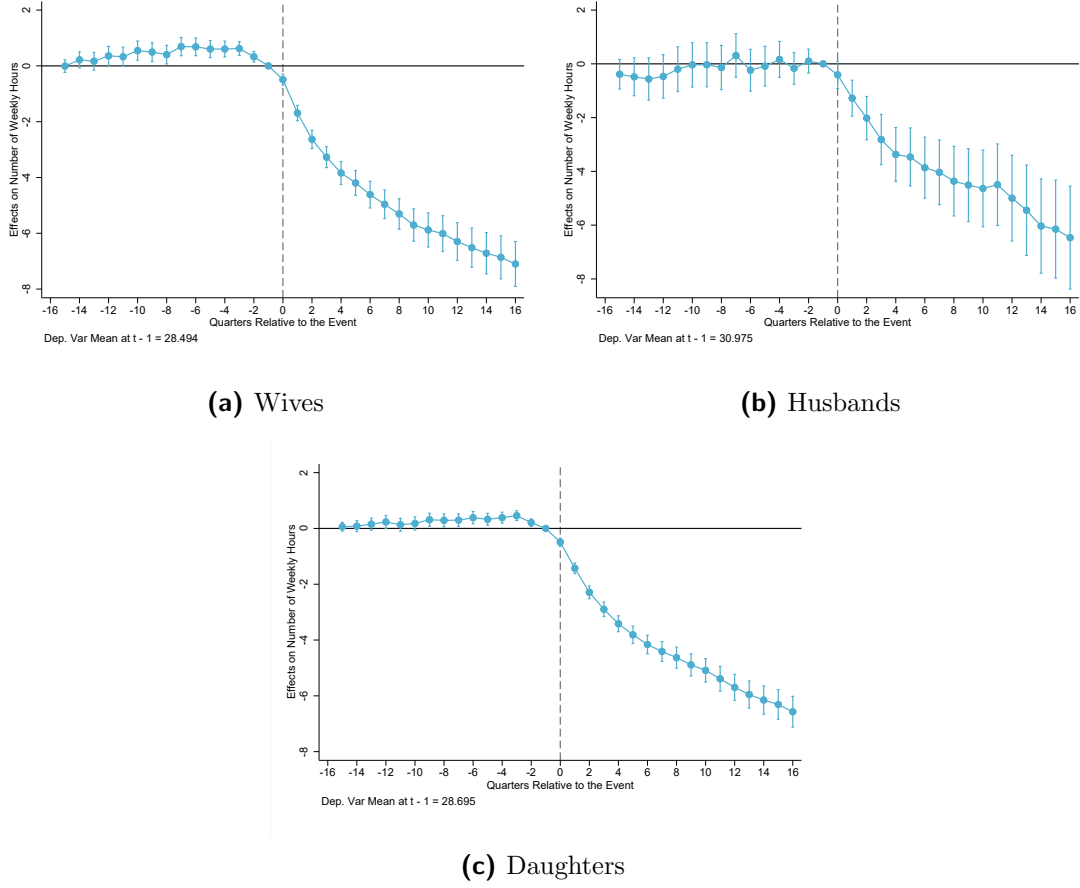


(a) Hours Worked

(b) Prob. of Employment

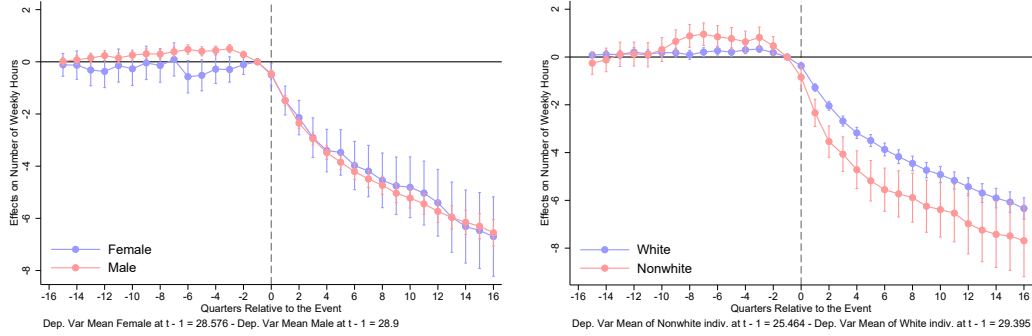
This Figure plots the average hours worked and probability of being employed relative to the event. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event.

Figure A3: Effects on Weekly Hours Worked - Relation with the Former Servant



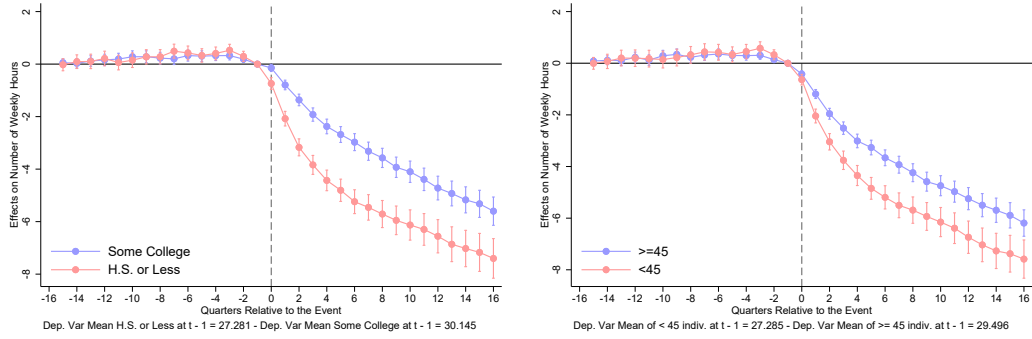
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event. Panel (b) sample is restricted to those employed in the given quarter. Standard errors are clustered at the individual level.

Figure A4: Effects on Hours - Heterogeneity Recipient Characteristics



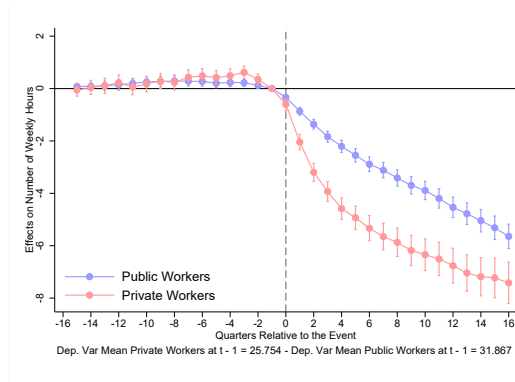
(a) Gender

(b) Race



(c) Education

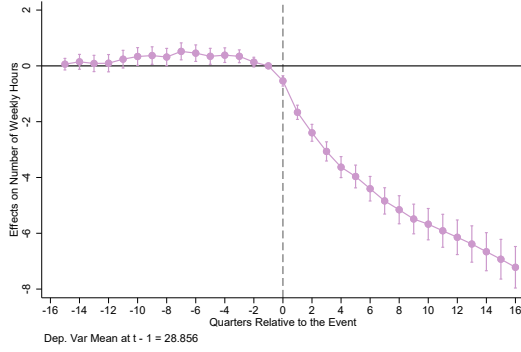
(d) Age



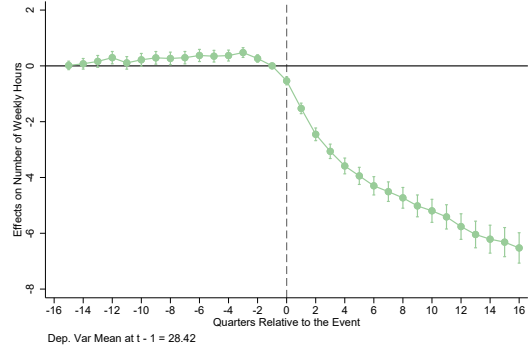
(e) Public vs Private Sector Employment

This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises all pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event. Each connected series restricts the sample of the event-study to the respective group described in the label.

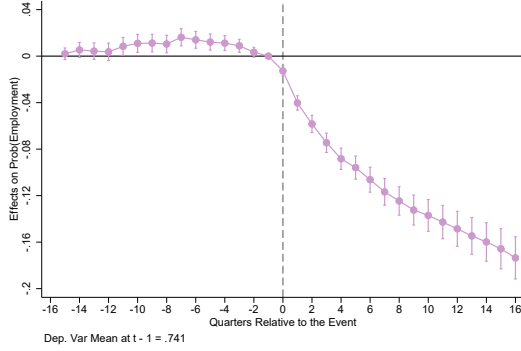
Figure A5: Effects by type of Pension - Executivo Federal vs Militares



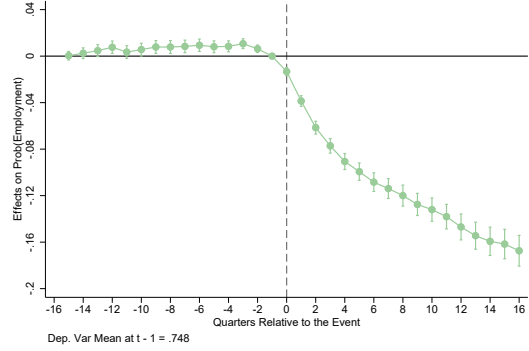
(a) Hours Worked - Executivo Federal



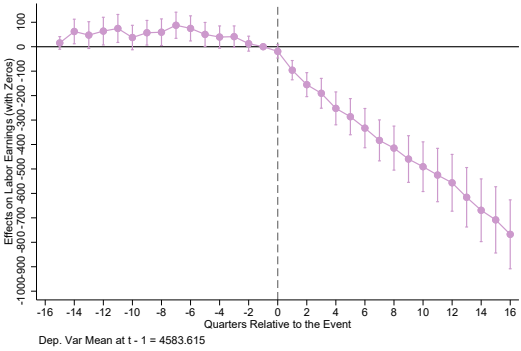
(b) Hours Worked - Militares



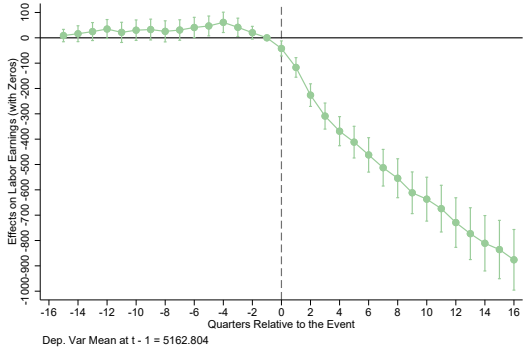
(c) Prob. (Employed) - Executivo Federal



(d) Prob. (Employed) - Militares



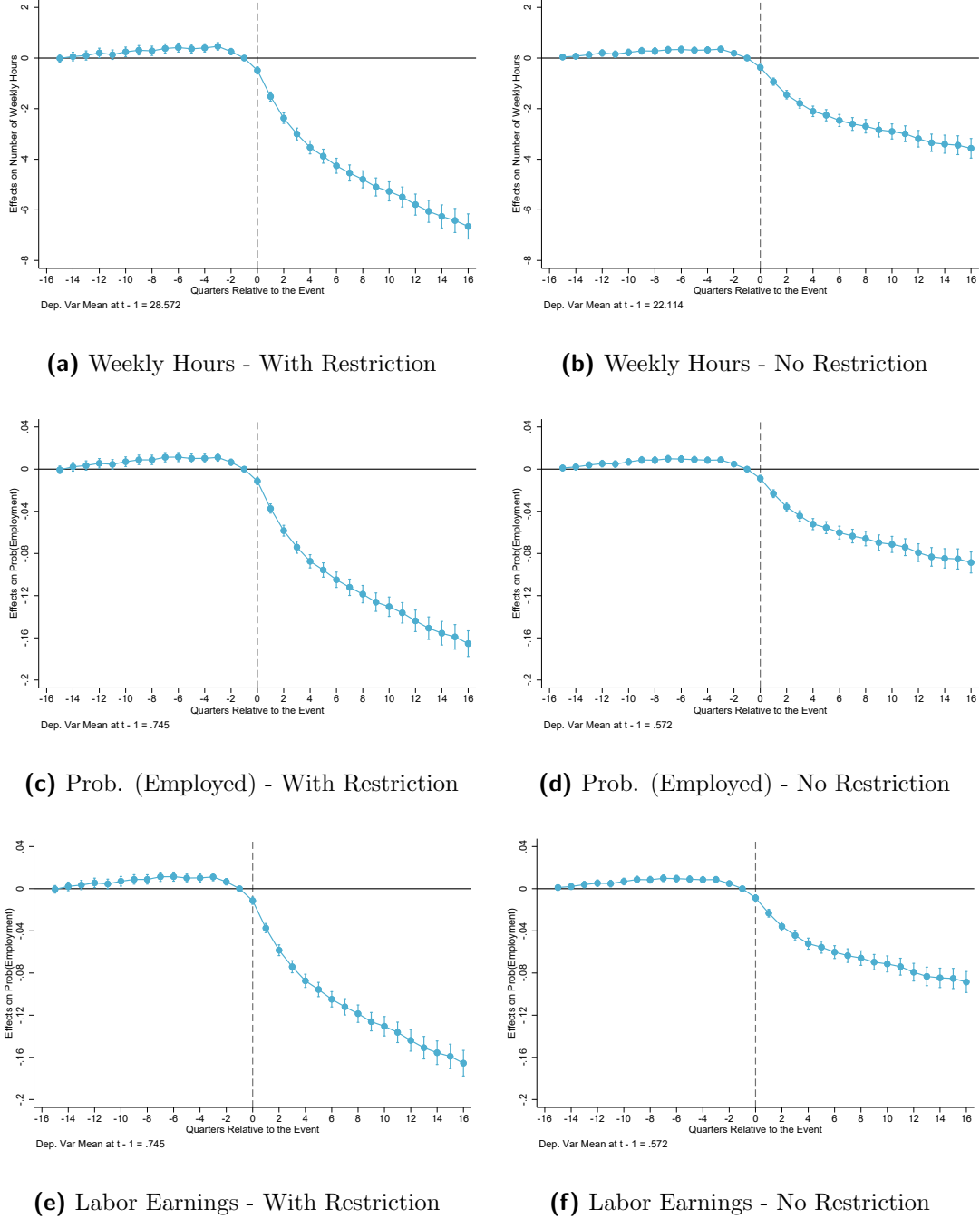
(e) Labor Earnings - Executivo Federal



(f) Labor Earnings - Militares

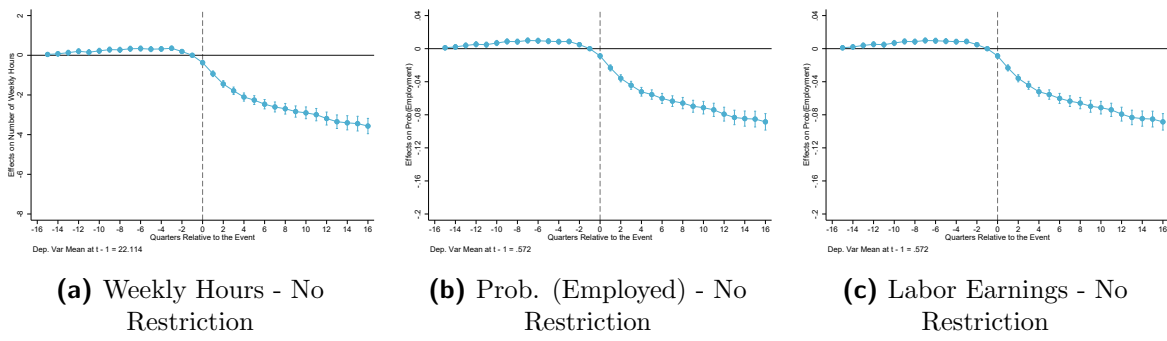
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises pension recipients between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event. Panels (a), (c), and (e) show estimates using a sample of recipients whose pensions originated from a former Federal Public Sector employee. Panels (b), (d), and (f) show estimates using a sample of recipients whose pensions originated from a former Military worker.

Figure A6: Effects on the Number of Hours Worked Without Previous Employment Restriction



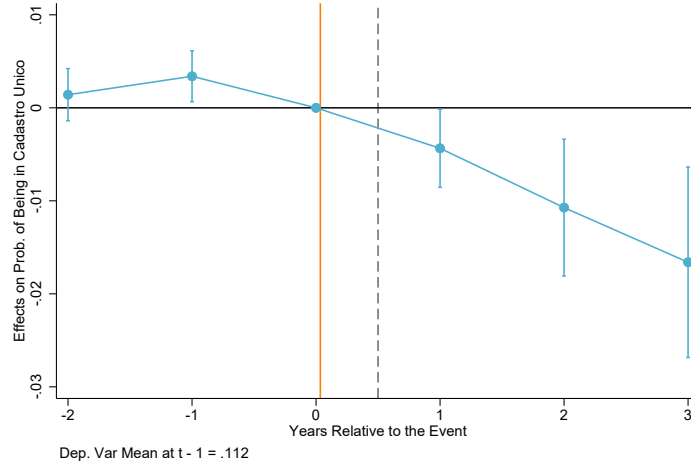
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises pension recipients between 25 and 55 years old at the time of the event. Panels (a) restrict the sample to recipients who had at least one formal job contract in the 4 years before the event. Panel (b) does not impose this restriction.

Figure A7: Effects on the Number of Hours Worked Without Previous Employment Restriction



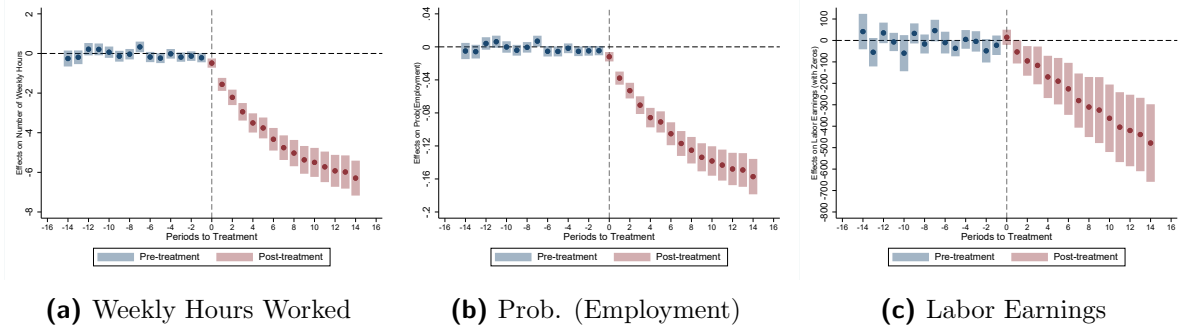
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises pension recipients between 24 and 55 years old at the time of the event. Panels (a) restrict the sample to recipients who had at least one formal job contract in the 4 years before the event. Panel (b) does not impose this restriction.

Figure A8: Effects on Being Enrolled in Cadastro Único



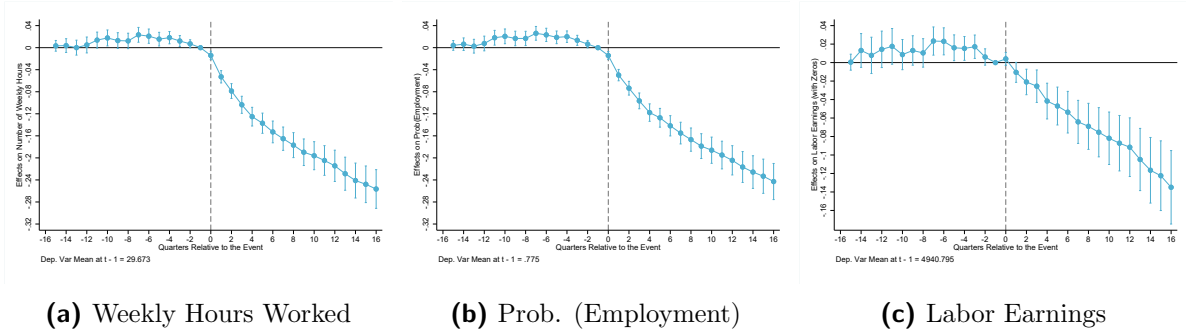
This Figure plots the estimates of coefficients β_k from a simplified version of equation 17 and their respective 95% confidence intervals. The sample comprises pension recipients between 25 and 55 years old at the time of the event. We restrict recipients to those who had events after 2014, as our Cadastro Único data is available only between 2012 and 2019. Due to the Structure of the outcome data, we use a Yearly Panel instead of a quarterly panel.

Figure A9: Sant'Anna and Zhao (2020) Estimates of Effects on Labor Supply



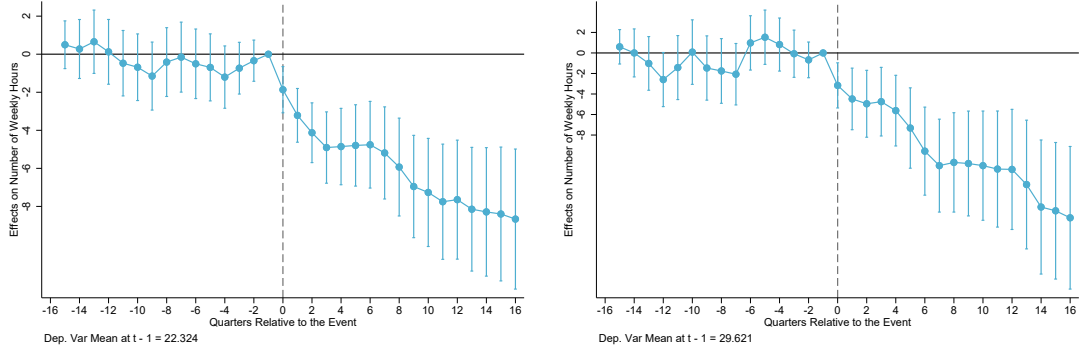
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals using Sant'Anna and Zhao (2020) estimators. The sample comprises all spouses between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event.

Figure A10: Poisson Estimates of Effects on Labor Supply



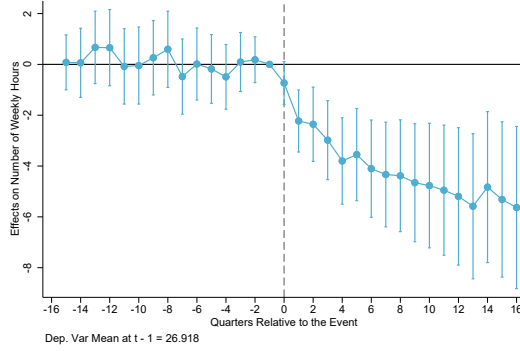
This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals using a Poisson Regression transformation to estimate the model. The sample comprises all spouses between 25 and 55 years old at the time of the event who had at least one formal job contract in the 4 years before the event.

Figure A11: Effects on Hours - Heterogeneity



(a) Sons

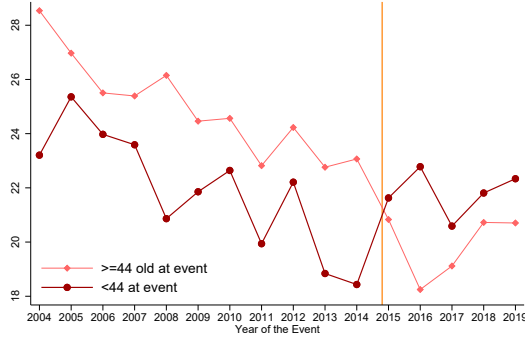
(b) Others - Men



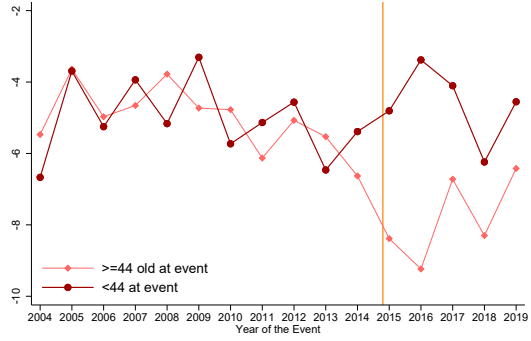
(c) Others - Women

This Figure plots the estimates of coefficients β_k from equation 17 and their respective 95% confidence intervals. The sample comprises pension recipients between 25 and 55 years old at the time of the event. Each Panel restricts the sample to the corresponding type of relation with the public servant.

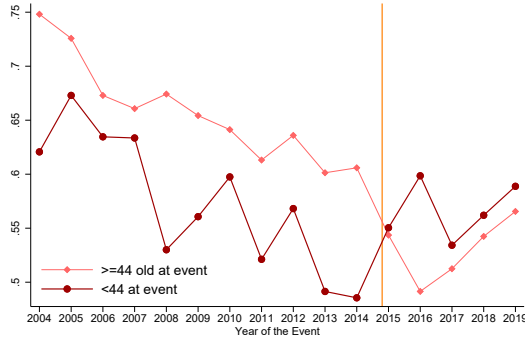
Figure A12: Average Labor Supply 2 Years After Event by Event Year



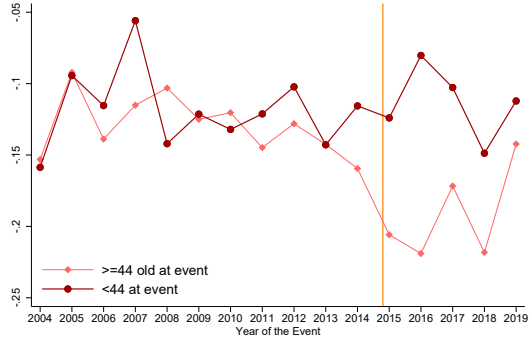
(a) Weekly Hours



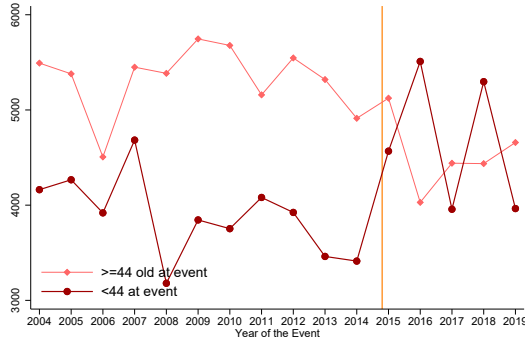
(b) Δ Weekly Hours



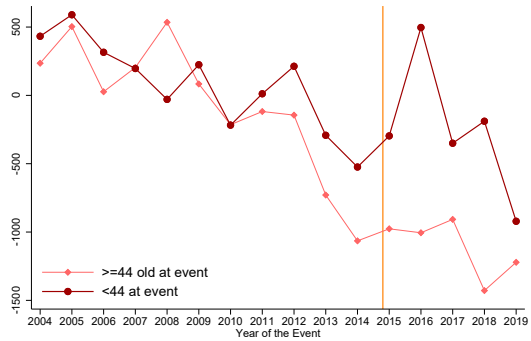
(c) Prob (Employed)



(d) Δ Prob (Employed)



(e) Labor Earnings



(f) Δ Labor Earnings

This Figure plots the average Labor Supply outcomes 2 years after the death of a spouse for pension recipients across event years. Δ represents $Y_i^{2years} - Y_i^{pre}$ where pre corresponds to the quarter before the event.

Table A2: Summary Statistics of Analysis Sample

	(1)	(2)	(3)	(4)	
	44 or Older		Younger than 44		All
	Before 2016	2016 or After	Before 2016	2016 or After	
Lifetime Pension	0.99	0.92	0.95	0.13	0.94
Avg. Age at Start	49.27	49.69	37.22	37.06	46.30
Sh. Female	0.81	0.83	0.83	0.80	0.82
Sh. Nonwhite	0.15	0.20	0.22	0.27	0.18
Sh. with High School	0.36	0.39	0.42	0.41	0.38
Sh. with College	0.39	0.41	0.39	0.48	0.40
Observations	5381	1091	1791	374	8637

This Table presents summary statistics of our sample to analyze the effects of the reform. 44 or older and younger than 44 refer to the age at the start of the pension

Table A3: Life Expectancy by Age in 2015

Age	All	Men	Women
30	48.1	45.1	50.9
35	43.5	40.7	46.2
40	38.9	36.3	41.4
45	34.5	32.0	36.8
50	30.2	27.9	32.3
55	26.0	23.9	28.0

This Table shows the life expectancy by age and gender calculated by the Brazilian Institute of Geography and Statistics (IBGE)

Table A5: Effects of Pensions' Duration on Labor Supply Two Years After Event

	Log(Benefits)	Weekly Hours		Prob(Employment)		Labor Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Levels	Δ	Levels	Δ	Levels	Δ
Treated x [2016,2018]	-0.0712 (0.06069)	-0.7740 (1.52484)	1.5030 (1.17153)	-0.0237 (0.03790)	0.0068 (0.02854)	-220.2578 (488.34413)	389.2681* (226.63747)
Treated x Post Reform	-0.3331*** (0.06449)	-0.7310 (1.55901)	1.9060 (1.17306)	-0.0250 (0.03895)	0.0096 (0.02879)	-632.2579 (489.37732)	208.2162 (222.27759)
Mean of Outcome (Treated)	9.44	26.55	26.55	.7	.7	5828.24	5828.24
Mean of Δ Outcome (Treated)			.36		.02		149.01
Observations	6612	6663	6663	6663	6663	6663	6663

This Table shows the estimates of β of equation 21 using labor supply 2 years after the event as the outcome. We include education, gender, and race dummies as controls, in addition to Fixed Effects of quarter of benefit start, and Age at Start. The sample comprises all spouses of former Executivo Federal and Military workers who started receiving benefits between 2016 and 2021. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Effects of Pensions' Duration on Labor Supply One Year Before the Event

	Prob(Lifetime Pension)	Weekly Hours		Prob(Employment)		Labor Earnings	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Levels	Δ	Levels	Δ	Levels	Δ
Younger 44 x [2004,2006]	-0.0173 (0.02260)	0.6065 (1.30949)	-0.6483 (1.20001)	0.0145 (0.03120)	-0.0134 (0.02918)	268.4042 (379.64830)	-124.6742 (181.51080)
Younger 44 x [2007,2009]	-0.0096 (0.02224)	-2.1473 (1.32898)	-2.6152** (1.18281)	-0.0409 (0.03211)	-0.0599** (0.02870)	266.9941 (412.38870)	-210.6960 (159.60475)
Younger 44 x [2010,2012]	-0.0106 (0.02321)	-0.6725 (1.35100)	-1.6828 (1.20569)	-0.0166 (0.03236)	-0.0440 (0.02902)	60.9558 (432.00211)	-180.6908 (163.02534)
Younger 44 x Post Reform	-0.7099*** (0.02232)	0.5127 (1.30960)	-1.3551 (1.15429)	0.0126 (0.03159)	-0.0403 (0.02773)	652.6221 (428.47295)	-178.3503 (166.99389)
Mean of Outcome (Treated)	.9	26.86	26.86	.68	.68	3932.69	3932.69
Mean of Δ Outcome (Treated)			2.32		.06		67.16
Observations	9499	9499	9499	9499	9499	9499	9499

This Table shows the estimates of β of equation 19 using labor supply 1 years before the event as the outcome. We include education, gender, and race dummies as controls, in addition to Fixed Effects of quarter of benefit start, and Age at Start. The sample comprises all spouses from former Executivo Federal workers who started receiving benefits between 2003 and 2018. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

10 Data Appendix

This appendix describes in detail the steps taken to construct the analysis sample from the raw administrative records. Section 2.2 describes the three main data sources; Section 4.2 provides an overview of the matching procedure and sample restrictions. Here we document the technical steps underlying each stage of the data construction, in the order in which they are executed.

B.1 Raw Pension Records: Structure and Standardization The pension data are obtained from the Portal da Transparência of the Brazilian federal government, which publishes monthly extracts of all active pension payment records. For the Executivo Federal

regime, the data span January 1994 through December 2022, yielding approximately 348 monthly files. Each file records one observation per active pension payment in that month. The files are distributed in CSV format with Portuguese-language variable names, which we standardize to a common schema upon import.

Each monthly record contains the following information on the beneficiary: full name, date of birth, digits 4 through 6 of the 11-digit CPF tax identifier, the nature of the relationship to the deceased servant (beneficiary type), and the gross and net monthly benefit amounts. It also contains the following information on the originating civil servant: full name, date of birth, date of death, government agency, occupation level, and digits 4 through 6 of the servant’s CPF. Additionally, the pension record includes the start and end dates of the benefit entitlement, the pension type (e.g., survivor pension under specific legislation), its legal nature (whether *vitalícia* or temporary), and the state (UF) where payments are processed.

A key limitation of the raw data is that the CPF is only partially disclosed. The portal masks the first three and last two digits of the 11-digit CPF, making it impossible to uniquely identify individuals across datasets using the CPF alone. The three digits we observe — positions 4, 5, and 6 — form what we refer to throughout as CPF6. This partial identifier is the foundation of the matching strategy described in Section B.3.

For the Militares regime, the data structure is similar but two fields are missing: the date of death of the originating military worker, and the date of birth of the beneficiary. As described in Section 4.2, this changes the matching strategy for military records.